



A structured approach to the design and testing of LOX/kerosene impinging-jet injectors

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ARTICLE INFO

Communicated by Dr Bing Wang

Keywords:

Liquid rocket engine
Injector
Impinging-jets
Cold-flow
Hot-fire
Design methodology

ABSTRACT

Liquid rocket engine injectors are complex components that significantly influence engine performance, combustion stability and manufacturing cost. They typically cycle through numerous iterations during development and achieving an injector design that successfully meets the target criteria often requires the application of prior experience, extensive simulation and testing, and ordinary trial and error. In an effort to reduce the heuristic nature of the process, this paper proposes a methodological approach to impinging-jet injector design for LOX/Kerosene engines. This structured approach considers the risks and limitations of early decisions, such as element and manifold types, and their downstream effects on the design process. As an illustration of the approach, the methodology is applied to the design of two different injectors with varying inputs for the same engine. Both of these injectors were manufactured and cold-flow tested to characterize fluid dynamic performance. Cold-flow tests were conducted with water as an analog to analyze the propellant distribution through each orifice and pressure drop. One injector was hot-fire tested in a technology demonstrator liquid rocket engine and performed nominally, with high c^* efficiency and stable combustion. The application of this methodology ensures a more deterministic approach to the injector design process potentially reducing the time and cost of development.

1. Introduction

Injectors are critical to ensuring efficient performance and stable combustion in liquid rocket engines. When designing an injector, consideration must be given to numerous aspects, including the propellant characteristics, propellant manifolds, material selection, engine integration, element design, propellant atomization and wall compatibility. As an added complexity, key decisions made early in the design process affect other design parameters and ultimately influence engine performance and stability [1,2].

Impinging jet elements for rocket engine injector applications have been studied since the 1950s [3–5] and continue to be a subject of interest today [6–11]. In parallel, studies on jet injection into high-speed crossflows have provided foundational insights into jet penetration, shock structures, and mixing characteristics [12]. These investigations form a critical baseline for understanding more complex jet configurations. Impinging jet dynamics are a key focus of research, including the formation of sheets, resultant ligament formation and sheet disintegration [13–15], primary and secondary droplet breakup [16–18], mixing and combustion [19,20].

The mechanisms involved in propellant injection, atomization, and combustion are well understood and are under continuous investigation. However, there is little guidance in the literature on structured approaches to balancing the trade-offs between certain design decisions in clean-sheet rocket engine design. Gill and Nurick [21] provide practical guidance on injector design with respect to considerations such as material selection, manufacturing methods and assembly constraints but do not offer a formalized methodology for injector design in its entirety.

This paper proposes a methodology that can serve as a structured guide in the clean-sheet design of liquid rocket engine propellant injectors. The methodology is limited to liquid oxygen (LOX) and kerosene injectors, as this is a commonly used propellant combination for booster stages of launch vehicles [22]. Impinging elements are chosen due to their ubiquity in LOX/kerosene engines versus coaxial elements. One of the goals of the methodology is to achieve uniform mixture ratio and mass flow distributions across the face of the injector, both of which are known to improve engine performance.

This systematic approach is detailed in Fig. 1. This is an iterative process, where many sub-processes feed back to one of the earlier steps. The methodology can be categorized into two distinct phases. The

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Nomenclature

Symbol description

A	area
A_e	area per element
A_n	area of annulus n
C_d	discharge coefficient
$\bar{D}$	droplet mass median diameter
d	orifice diameter
E_m	mixing factor
E_n	number of elements in ring n
Fe	Ferrenberg performance approximation
G	width of annular ring
K	head loss coefficient
K_v	flow coefficient
L/d	length-to-diameter ratio
$\dot{m}$	mass flow rate
n	ring number
$N_{orifices}$	number of orifices
ΔP	pressure drop
ρ	density
v	velocity

Abbreviations

ABLE	ablative blowdown liquid engine
ASRI	aerospace systems research institute
EBW	electron beam welding
EDM	electrical discharge machining
FFT	fast fourier transform
HRHP	high-risk-high-performance
LRLP	low-risk-low-performance
O-F-O	oxidizer-fuel-oxidizer
PSD	power spectral density
SAFFIRE	South African first rocket engine

design phase contains the majority of the decisions and yields an initial geometry for numerical analysis. A simplified structural analysis is required to confirm the suitability of the design before proceeding to the more in-depth computational fluid dynamics (CFD) analysis.

The inputs to the methodology are engine and chamber specifications (pressures, flow rates and chamber diameter), risk-tolerance, manufacturing methods and material selection. Risk-tolerance is a binary high-risk-high-performance (HRHP) or low-risk-low-performance (LRLP) input. This arises from the premise that parameters that improve mixing and combustion efficiency typically have a corresponding negative impact on combustion stability [1] such that engine performance and risk are positively associated. This input serves to inform the decision between options in a qualitative manner, where the quantifiable impact on engine performance and stability cannot be known early in the design process. A specific example is the choice between a like-doublet or unlike-triplet element type, where the doublet is inherently more stable than the triplet [21], but has a lower mixing efficiency [7].

The element design process is driven by the required pressure drop [21], impingement distance [23,24], length-to-diameter ratio [25], mixing efficiency [26] and element pattern spacing requirements. The proposed element patterning method determines the element layout and uses a geometric correlation to distribute elements uniformly across the faceplate. Manufacturing methods and material selection significantly impact the manifold design of an injector. Manifold type, wall thickness, and space for joints, welds and seals all influence the element spacing, pattern and orifice geometry. Therefore, the criteria in the manifold design sub-process serve to reinitialize or exit the injector design loop.

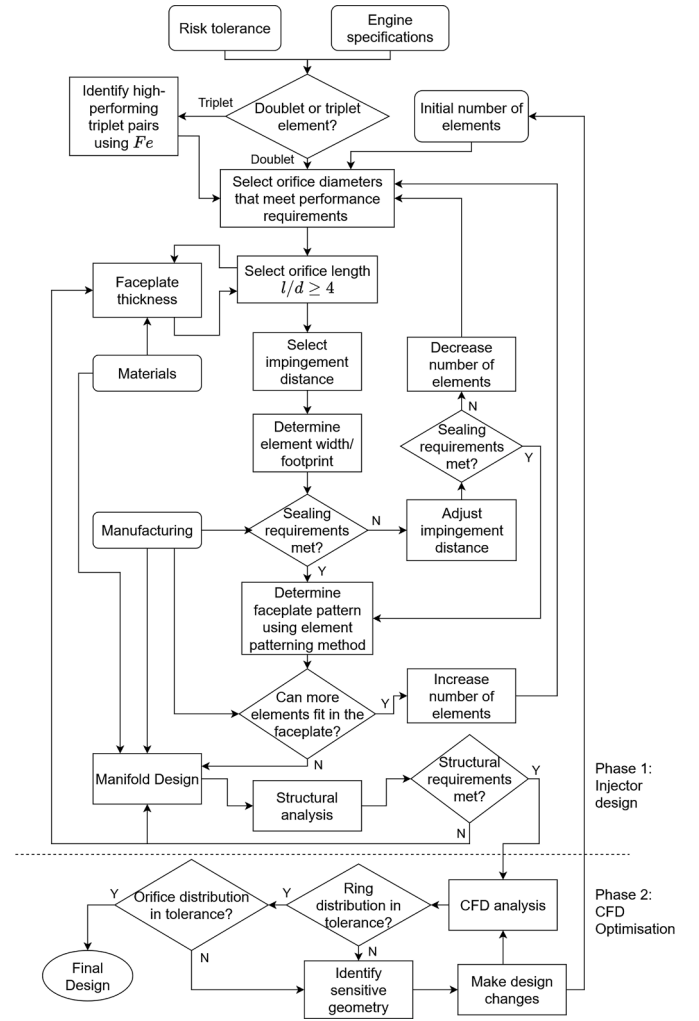


Fig. 1. Design methodology flowchart.

The injector manifold is then analyzed through a CFD approach to determine both grouped and individual orifice flow rates. If an orifice's flow rate is not within the targeted tolerance of the design condition, the manifold geometry is adjusted iteratively. The general approach is thus to use analytical and empirical methods for element design and spacing in conjunction with CFD to ensure orifices are supplied with the appropriate inlet conditions to match the inputs to the analytical design.

Relying on empirical data for element design enables the relatively rapid evaluation of multiple injector configurations. These configurations can then serve as candidates for more detailed investigation, such as high-fidelity CFD simulations and experimental testing, which fall outside the scope of the current methodology. Such follow-on studies are typically required to characterize the impingement dynamics, spray interaction, and combustion behavior resulting from element pattern intermixing.

The proposed methodology is then employed to design and test two different injectors for the 18 kN-thrust Ablative Blowdown Liquid Engine or ABLE, a LOX/JET A-1 technology demonstrator rocket engine. A comparison is made between the cold-flow and hot-fire performance of the injectors and the expected performance as per the design methodology. The engine specifications for ABLE are shown in Table 1. The two injector designs share the same engine specification input, but use different risk-tolerance and manufacturing inputs for the design methodology.

Table 1
ABLE specifications.

Parameter	Units	Value
Thrust (ideal)	kN	18
Chamber pressure	bar	12
Chamber diameter	mm	220
LOX mass flow	kg/s	5.66
JET A-1 mass flow	kg/s	2.27
Film-cooling fraction	%	10

The methodology proposed here is motivated by two related factors. The first is a global surge in launch vehicle development programs, many of which involve LOX/kerosene engine choices for which this methodology is useful. The second is the location of many of these programs in countries with small aerospace sectors where engineers lack the legacy design experience that enables heuristic interventions to mitigate risk and accelerate project timelines. In this sense, the methodology may be seen as a first attempt to codify the engineering lore of impinging injector design.

The need for a clean-sheet injector design methodology was realized during the development of the South African First Rocket Engine (SAFFIRE), a small-satellite launch vehicle engine currently in work at the University of KwaZulu-Natal's Aerospace Systems Research Institute (ASRI). The program's objective is to develop a LOX/kerosene electropump engine to be used in a clustered configuration for a commercial launch vehicle able to deliver a 200 kg payload to a 500 km orbit from Africa [27,28]. The methodology was developed during the design of SAFFIRE's forerunner, ABLE, which served as a prototype battleship engine for the experimental validation of the design approach, the liquid rocket engine test facility and the test operation [29]. The methodology is intended to reduce development time and cost by emphasizing best practices from historical programs and by guiding the process towards sound design decisions that mitigate risk in costly engine development programs.

2. Injector design methodology

This section details the inputs to the methodology and the sub-processes within the injector design loop, as shown in Fig. 1. Following this iterative process will yield an initial injector design that meets the desired pressure drop, atomization, stability, and structural requirements and is suitable for further investigation. The design is then used as the initial condition for the CFD optimization loop.

2.1. Inputs

Inputs to the design methodology are engine specifications, manufacturing methods, material and risk-tolerance. Engine specifications such as chamber pressure, flow rate, chamber diameter and engine integration all affect the injector design. The element pressure drop required in the methodology is a function of the chamber pressure. The flow rate, combined with the required pressure drop, is an input to element shape and pattern. The chamber diameter affects the available space on the faceplate for element patterning.

Manufacturing methods and materials both have a direct influence on the manifold geometry. Manufacturing constraints on the manifold limit the possible element design and element pattern configurations, while materials affect manifold geometry through their structural properties and compatibility with specific manufacturing methods. Advanced manufacturing technologies such as additive manufacturing greatly assist in obtaining a manifold geometry with the desired element design and element pattern, but these technologies are not always well-established in developing countries.

The impact of specific design decisions on an injector's performance and stability is often uncertain during the design phase. Poor design

choices can lead to combustion instability, marked by uncontrolled pressure fluctuations that may damage the combustion chamber and destroy the engine [1]. To address this, the proposed methodology aims to mitigate such risks, with a brief overview provided on the causes and analysis of combustion instability.

Sensitive time lag theory, as described by Crocco and Cheng [30], remains a common framework for understanding combustion instability [31]. This theory assumes combustion is a binary event, with time and space lags representing the delay and distance between propellant injection and combustion onset. Instabilities are more likely when the oscillation period and the combustion's characteristic time are of the same order of magnitude. These properties are influenced by chamber and injector design, allowing for some early design interventions.

Throughout combustion, multiple mechanisms can couple with chamber acoustics, causing instability. These include propellant injection, atomization, droplet heating, vaporization, mixing, and chemical reactions. However, at subcritical conditions, vaporization typically controls the combustion rate due to its longer characteristic time [31]. Primary atomization indirectly influences this by affecting droplet size and vaporization rate. If the rate-controlling process has a sufficiently long characteristic time, it may be less influenced by chamber acoustic oscillations. Larger engines are particularly prone to instability since their acoustic frequencies are more likely to align with these mechanisms.

While numerical stability analyses using high-fidelity CFD models [32,33] are available, they are not practical for early design stages. Instead, the proposed approach focuses on understanding how rate-controlling mechanisms influence instability. Designers should assess whether a design change affects the rate-controlling process; if it decreases this time, it might enhance performance but reduce stability, making it a HRHP choice.

2.2. Element type

The initial step in the methodology is selecting the injection element, which influences manifold design and manufacturing requirements. This choice depends on the propellant combination, O/F ratio, propellant phase, chamber pressure, pressure drop, and available manufacturing techniques. For LOX/kerosene engines with a typical O/F ratio of 2.6 [34], like-doublets and unlike-triplets in an oxidizer-fuel-oxidizer (O-F-O) configuration are suitable. Unlike-doublets reduce atomization efficiency due to differing jet diameters and momenta [3]. Quadruplets and pentads add complexity without significant performance benefits.

When deciding between a like-doublet or unlike-triplet, the primary driver is the risk-tolerance of combustion instability. Like-doublets are inherently more stable than unlike-triplets, according to Gill and Nurick [21], although the authors provided no supporting reasons. From the concepts outlined in the sensitive time lag theory and rate-controlling mechanism, there are several possible explanations for the stability difference:

1. Droplet formation in unlike-triplet elements is influenced by the velocity profiles of the fuel and oxidizer jets, which react differently to changes in chamber pressure due to differences in propellant properties, orifice geometry, and manifold pressures. Variations in chamber pressure can alter each jet differently, affecting impingement and droplet formation and ultimately impacting the rate-controlling mechanism and potential coupling.
2. Better mixing produced by the triplet will result in faster droplet heating and vaporization, shortening the characteristic time of the rate-controlling mechanism and increasing the probability of coupling with the high-frequency acoustic oscillations in the chamber.
3. The slower mixing of the like-doublet and variation in the axial velocity components and droplet size for each propellant will result in a larger combustion region further downstream. As per Crocco and Cheng [30], spreading the combustion zone across the chamber

Table 2
Propellant properties [38–40].

Property	Units	LOX	JET A-1
Density	kg/m ³	1142	807
Viscosity	μPa s	0.2	1.92
Critical pressure	MPa	5.04	23.44
Critical temperature	°C	−118.5	410

reduces the influence of anti-nodal surfaces on the combustion process. This also has the stabilizing effect referred to as impinging jet displacement sensitivity [35].

Chamber pressure selection must be considered when selecting an injection element based on risk-tolerance, especially as it approaches or exceeds the supercritical pressure of propellants [36]. In the supercritical state, fluids exhibit zero surface tension, combining liquid density with gas-like viscosity [37]. This state enhances atomization and reduces vaporization time, increasing the likelihood of coupling with high-frequency acoustic oscillations, thereby raising instability risks.

However, this does not necessarily mean that chamber pressures below a propellant's supercritical pressure are free of fluids in a supercritical or transcritical state. It is possible that in chambers near supercritical pressure, pressure waves could momentarily cause the propellant to enter that state. This provides an opportunity for coupling as the time of the rate-controlling mechanism can be impacted by pressure disturbances in the engine. The supercritical points and other relevant properties of LOX and JET-A1 are shown in Table 2. Commercially available JET A-1 can have a high variation in density. A median value of 807 kg/m³ is used for the calculations and simulations, where the actual density is measured before tests.

2.3. Element Design

Following selection of either a like-doublet or unlike-triplet element the orifice geometry can be designed. This process is iterative, as it depends on the number of elements defined by the faceplate pattern, which itself is influenced by the chosen element geometry (see Fig. 1). The details of element quantity and distribution are covered in Section 2.4.

The injection element's performance is linked to the mixing factor, E_m , introduced by Rupe [3], where $E_m = 1$ indicates perfect mixing and $E_m = 0$ indicates no mixing. Originally developed for unlike-doublets, E_m describes mixing across various injector types, with numerous studies identifying conditions to maximize E_m [7,41]. This methodology leverages existing experimental data to optimize parameters that enhance E_m .

Key design factors such as orifice geometry, impingement distance, and angle apply to both element types. While the orifice L/d ratio does not significantly impact E_m [42], it must be large enough to prevent hydraulic flip (flow separation in the orifice), with a minimum L/d of 4 being recommended [25]. Studies by Ferrenberg et al. [26], Hoehn et al. [43] show that the impingement angle minimally affects E_m , though larger angles can cause backslash. A total impingement angle of 60° is widely studied and favored for optimal mixing [5,21,43], making it the recommended choice in this design approach.

Impingement distance, measured from the orifice outlet to the meeting point of propellant streams, is typically set at an impingement distance ratio of around 5 [24], but should not exceed 7 [21] to minimize the risk of misaligned jet impingement resulting from orifice manufacturing tolerances. Smaller ratios are favored as they reduce orifice spacing, allowing for denser faceplate patterning.

Orifice inlets can be rounded, chamfered, or sharp-edged, with increasing pressure drops in that order. Despite reduced pressure drop with rounded inlets, sharp-edged inlets are preferred due to simpler manufacturing and more consistent and predictable pressure drops [21]. While sharp edges can lead to burr formation, affecting the discharge coefficient (C_d), this can be minimized through deburring and polishing.

Table 3
Range of tested parameters for Eq. (2) [26].

Parameter	Units	Min	Max
Velocity ratio	–	0.37	1.69
Diameter ratio	–	0.92	1.58
Density ratio	–	0.76	1.32
Impingement angle	°	30	90
Orifice diameter	mm	1.17	2.36
Injection velocity	m/s	6.70	20.20

Electrical discharge machining (EDM) or wire-cutting is recommended to prevent burrs entirely.

A minimum pressure drop is necessary to decouple feed system pressure fluctuations from the combustion chamber [1]. For unlike impinging elements, a pressure drop of 10% to 15% of chamber pressure is sufficient, while like impinging elements require 15% to 25% [21]. Using Eq. (1) adapted from Huzel and Huang [1], the pressure drop can be calculated, with a head loss coefficient K of 1.7 for sharp-edged orifices. This methodology mandates a minimum pressure drop of 15% of the chamber pressure for either element type.

$$\Delta P = \frac{0.811 K \dot{m}^2}{\rho d^4 N_{orifices}^2} \quad (1)$$

2.3.1. Unlike-triplet

Ferrenberg et al. [26] conducted a study on the mixing performance of triplet elements with varying liquids, jet velocities and jet diameters. The mixing performance of a triplet element can be quantified by the empirical formula in Eq. (2) and is referred to in this article as the Ferrenberg performance approximation, Fe . The unlike-triplet is in the O-F-O configuration, meaning that the angled outer orifices are for the oxidizer while the vertical center orifice is for the fuel. The parameters used in Eq. (2) are labeled, such that the subscripts o and f refer to the oxidizer and fuel orifices, respectively.

$$Fe = \left(\frac{d_o}{d_f} \right)^{0.29} \left(\frac{\rho_o}{\rho_f} \right)^{0.56} \left(\frac{v_o}{v_f} \right)^{0.25} \quad (2)$$

Maximizing Fe in Eq. (2) maximizes the mixing coefficient, E_m . The recommended ranges of parameters to be used in Eq. (2) are given in Table 3 and were derived from the tests conducted by Ferrenberg et al. [26]. The maximum and minimum values of the recommended ranges serve as limits for the respective parameters in this triplet design methodology.

As per Table 3, the minimum orifice diameter is either limited to 1.17 mm or to a diameter that results in a velocity of 20.2 m/s or more. Similarly, for the maximum orifice diameter/minimum velocity. The density ratio is a function of the propellant combination and is therefore constant at approximately 1.4 for LOX and kerosene (temperature dependent). Fig. 2 shows the process for iterating through every possible diameter combination that meets the constraints detailed in Table 3.

2.3.2. Like-doublet

The like-doublet is a suitable choice for engine designs that have a low risk-tolerance for combustion instability, such as those with large combustion chambers and relatively low characteristic frequencies. This is due both to its inherent stability and that it enables control of propellant injection velocities.

This element does not directly mix propellant but instead relies on inter-element mixing. Therefore, the mixing performance of the doublet is dependent on the number of elements and their positioning relative to each other. As a result, the primary driver for like-doublet design is the element pattern (number and placement of elements) and the minimum pressure drop requirement, specified to be 15% for this methodology, while taking into consideration jet velocities and relative propellant axial velocities.

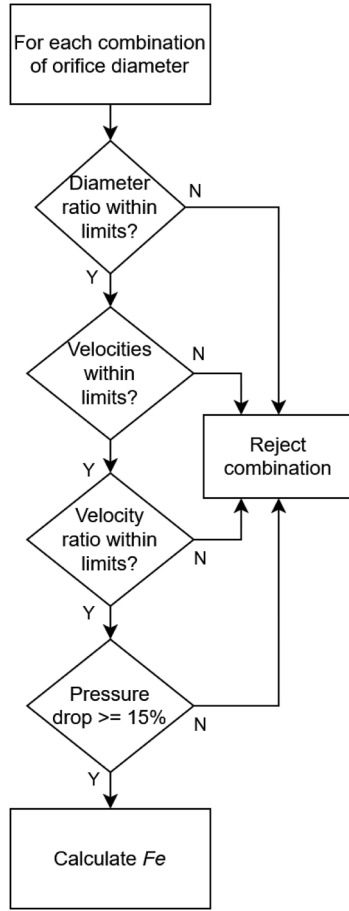


Fig. 2. Process to identify suitable triplet combinations.

The droplet mass median diameter for a like-doublet injector with a total impingement angle of 120° (60° half-angle) can be estimated using the empirical relation given in Eq. (3) [44]. Since a 30° half-angle is recommended for this methodology, the correction factor in Eq. (4) should be applied in conjunction with Eq. (3) to estimate the expected droplet mass median diameter [44]. Here, $\bar{D}$ is the droplet mass median diameter in μm for a given impingement half angle, d is the orifice diameter in mm , v is the jet velocity in m/s , and θ is the half angle in degrees.

$$\bar{D}_{60^\circ} = 4.5 \times 10^3 \frac{d^{0.57}}{v^{0.85}} \quad (3)$$

$$\bar{D}_\theta = (1.44 - 0.00734\theta)\bar{D}_{60^\circ} \quad (4)$$

An effective strategy for reducing the probability of transverse combustion instability with like-doublet elements is to reduce the velocity of the fuel jets. This has several effects, primarily increasing the droplet size and therefore vaporization time. Not only does it have a positive effect by increasing the characteristic time of the rate-controlling mechanism, but it displaces the combustion zone further downstream in the combustion chamber [45]. When the combustion zone is located near the faceplate, where the oxygen vapor is not sufficiently uniform, a spinning mode instability is more likely to be formed and maintained [35]. Moving the primary combustion zone further downstream, where the oxygen vapor is more uniformly distributed, mitigates this risk. An oxidizer-to-fuel velocity ratio of 3:1 is considered to have the most impact on stability [35].

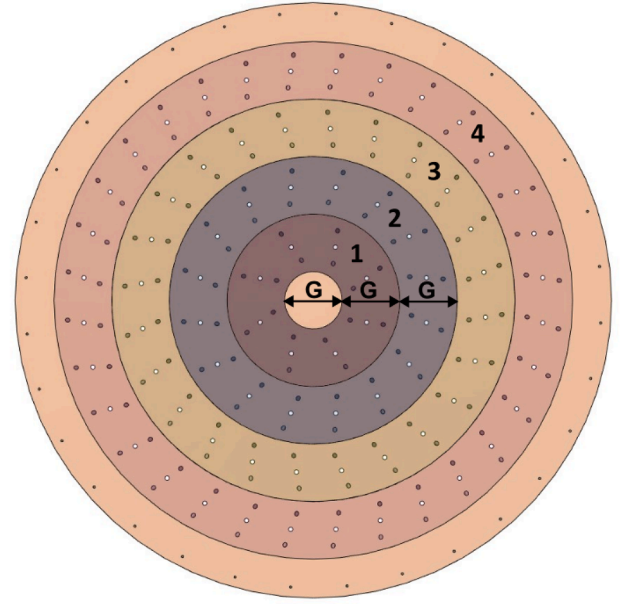


Fig. 3. Annular rings (1–4) of constant width G on the injector faceplate.

2.4. Element distribution

The element distribution pattern and therefore the number of elements determines the required orifice diameter to meet the minimum pressure drop. The element pattern is constrained by the manifold design which is often constrained by the selected manufacturing processes. One of the primary goals of this methodology is to maximize the uniform mixture ratio distribution and mass flow distribution and therefore maximize the number of elements on the faceplate. The maximum number of elements on a faceplate is limited by the minimum orifice diameter. As the flow rate through each orifice decreases, the diameter must be reduced to maintain the required pressure drop. The minimum diameter is either limited by element design constraints, as is the case with the triplet, or manufacturing constraints.

Depending on wall compatibility requirements, adjusting the mixture ratio near the chamber walls such that it is fuel-rich may be necessary. This is known as O/F zoning [46] and can be achieved via dedicated film-cooling elements. Element pattern and orientation can also be adjusted near the wall to achieve a lower O/F ratio. O/F zoning or film-cooling requirements are generally set as a specification for a particular chamber design.

The annular patterning method determines the element arrangement on the injector faceplate using a geometric relationship between annulus thickness and diameter. The faceplate is divided into concentric annular regions with a constant thickness G , equal to the inner diameter of the first ring. Each ring is numbered n based on its distance from the center, starting with 1 for the innermost ring. This method results in a linear relationship between the annulus area A_n and the ring number n , as shown in Eq. (5). Fig. 3 illustrates an example with four rings and film-cooling elements along the perimeter.

$$A_n = n2\pi G^2 \quad (5)$$

The faceplate area per element A_e for the first ring is calculated using annulus thickness G and the number of elements E_1 (Eq. (6)). Since the ring area depends on the ring number (Eq. (5)), maintaining a constant A_e as ring 1 means the element count in subsequent rings depends on G and E_1 (Eq. (7)). Maximizing elements in the first, most constrained ring optimizes the overall pattern while ensuring uniform distribution. This approach also simplifies manifolding by aligning features, such as

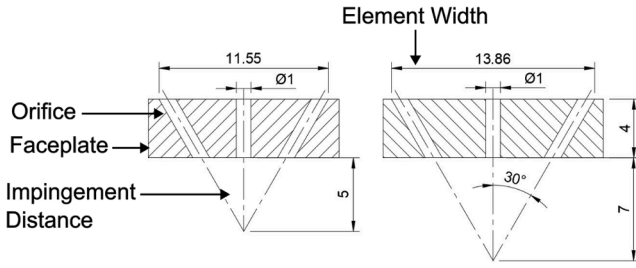


Fig. 4. Effect of impingement distance on the overall width of an element.

radial passages, with the injection element pattern.

$$A_e = \frac{2\pi G^2}{E_1} \quad (6)$$

$$E_n = nE_1 \quad (7)$$

The impingement distance has a direct effect on the element pattern. Fig. 4 shows the effect of varying impingement distance on the element width. Larger impingement distances require more space between orifice inlets. This affects the surrounding manifold and minimum annular width, G , to allow for the larger inlet requirements. Therefore, the impingement distance also directly affects the total faceplate pattern.

Maintaining a constant A_e for each ring is not always feasible in the face of conflicting requirements, for example manifolding, sealing and element geometry. In these cases, A_e can be used as a metric for element distribution uniformity. Element orientation relative to each other and the walls is an important consideration. Spray fans can be positioned to intersect and promote mixing or diverge and reduce mixing. Similarly, they can be orientated to avoid high concentrations of propellants near the chamber wall to avoid localized heating and burn-through.

2.5. Manifolds and manufacturing

Manifolds that enable high-density element patterns are complex and can be expensive to manufacture. The type of element, manifold, materials and manufacturing methods are all constraints when attempting to achieve a specific element pattern.

2.5.1. Material

Materials for LOX/kerosene engines must be chemically compatible with propellants and possess suitable mechanical properties at expected temperatures. Key constraints include oxygen compatibility and mechanical performance at cryogenic temperatures. Kerosene is generally non-reactive, so a wide range of aerospace-grade materials is suitable for its manifold, though elevated temperatures in regeneratively cooled engines must be considered. Common materials with the required oxygen compatibility and cryogenic toughness include stainless steel (300 series), nickel alloys (Inconel X750, 718), copper alloys, brass, Monel, and silver [21,47].

Titanium and aluminum have suitable properties at cryogenic temperatures and favorable densities but are not considered safe for injector design due to their reactivity with oxygen in the absence of their oxide layer [47]. The 300 series stainless steels meet the chemical and temperature requirements for LOX/kerosene injectors, are less expensive than alloys such as Inconel, and are thus recommended for injector body material in this methodology.

The faceplate experiences the highest temperature in the injector. This increases the mechanical demand of the faceplate material at elevated temperatures and the corrosion resistance requirements. Corrosion due to increased temperatures can result in the change of geometry of elements and therefore affect the performance and stability of an engine as well as potentially leading to structural problems [21].

Heat flux to the injector faceplate depends on element design, pattern, and propellant properties. There is no analytical model available in



Fig. 5. BPM5 integral injector design by Copenhagen Suborbitals [54].

the literature to determine this analytically. Radiation from the combustion zone is generally ignored, assuming the propellant cloud acts as a reflective barrier [48]. Thus, convective heat transfer, influenced by gas velocity and temperature, dominates. Faceplate temperature and heat flux are typically measured using thermocouples. Boylan et al. [49] reported gas velocities of 170–260 m/s and temperatures of 1700–1950°C for a LOX/Heptane engine with doublet elements. However, predicting these values in the design phase is challenging due to combustion zone recirculation effects.

While numerical methods effectively predict heat transfer to chamber walls and throats [50–52], applying them to faceplate heating is challenging. This is especially true for liquid propellants due to complexities in modeling injection, atomization, and combustion. Thus, this methodology avoids such simulations and instead opts for a conservative faceplate material selection approach.

The recommended choice for this methodology is to use either copper or high-strength nickel alloy such as Inconel 718 as the faceplate material. Inconel has been successfully used in many high-temperature applications due to its high melting point and favorable mechanical properties at elevated temperatures [53]. The choice between copper or Inconel is determined by structural requirements, budget and availability.

2.5.2. Manifolds

Multiple types of manifolding exist for rocket engine injectors, with common arrangements being integral injectors and concentric ring injectors. An integral injector is a single, solid body with the elements and manifolding features machined into it, as illustrated by Fig. 5. This simplifies the sealing requirements and reduces the number of components, reducing complexity and cost.

Integral injectors face challenges due to limited manifolding options inherent in traditional manufacturing. Common designs rely on drilled passages to supply orifice inlets, which restrict the element pattern and may leave burrs in inaccessible areas. To address this, EDM drilling is recommended.

Recent advances in additive manufacturing (AM) have enabled more complex integral injector designs that were previously impractical. AM allows integration of intricate geometries with rapid production, even using pre-processed components as the build plate.

Since integral injectors are made from a single material, it must satisfy both manifold and faceplate requirements. Limited heat flux predictability often restricts material choices to copper or Inconel, but copper's low strength may be unsuitable for high-pressure engines. AM could overcome this by joining dissimilar metals [55], as demonstrated with combinations such as 316 stainless steel with Inconel 718 [56] and 316L stainless steel with copper [57].

The concentric ring injector design, though more complex to manufacture and assemble, supports a high-density element pattern. The faceplate is composed of concentric rings, brazed or welded to the injector body (see Fig. 6) and is a natural fit with the annular patterning method. Using dissimilar materials, typically stainless steel for the body and copper for the faceplate, optimizes mechanical properties and cost. To prevent misalignment, it is recommended to drill the faceplate as a single unit before assembly, as misaligned orifices can degrade engine performance and stability [10].

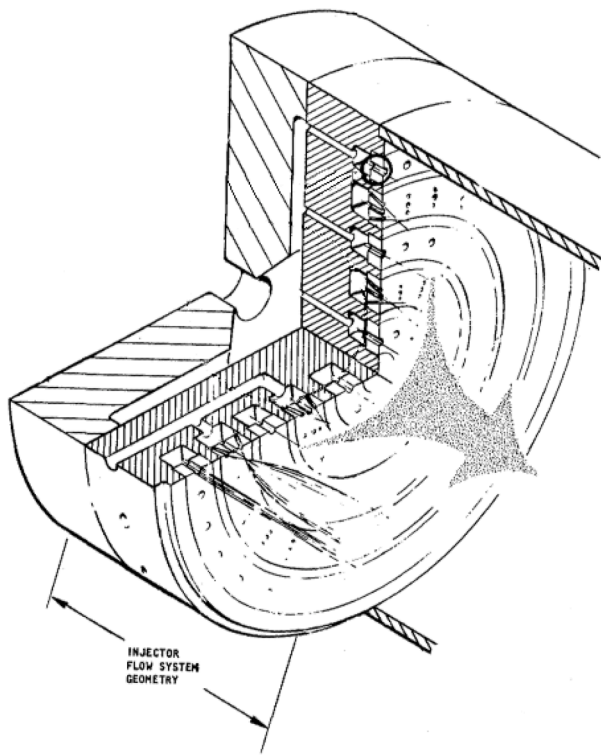


Fig. 6. Concentric ring injector, adapted from [21].

2.5.3. Manufacturing

Manufacturing methods must consider constraints such as chamber diameter and interfaces, which limit the space for propellant flow domains and element geometry. Welding techniques such as electron beam welding (EBW) and brazing are preferred for joining dissimilar metals. However, brazing can anneal cold-worked copper, reducing its strength. Silver-based solder is recommended due to its compatibility with oxygen and cryogenic resilience. Despite the relatively high melting point of the solder, implementing brazed joints may reduce the overall operating temperature of the faceplate.

Proper sealing is crucial to prevent leaks between fuel and oxidizer domains, which could lead to catastrophic engine failure. Double seals with a dedicated vent path can minimize this risk (see Fig. 7), though they require extra space, reducing element density and potentially impacting performance. Mechanical seals (such as O-rings and gaskets) are challenging to implement in such compact configurations due to their spatial requirements. The LRLP approach emphasizes safety but sacrifices inter-element mixing efficiency and overall performance.

The injector serves as the structural interface between the combustion chamber and gimbal, withstanding both engine thrust and internal pressure. High chamber pressures increase the injector pressure and the pressure differential ΔP across the faceplate. However, the highest mechanical loads occur during engine start-up, shutdown, or cold-flow testing when there is no back-pressure from the chamber and ΔP is highest. Thus, the injector must be designed to survive maximum internal pressure without relieving forces [58]. Cold-flow testing the injector at operating pressures and flow rates is likely to induce the most mechanical stress on the injector structure.

Propellant inlet positioning is often constrained by engine and plumbing layouts. Ideally, inlets should not disrupt flow distribution in the manifold. If the design injector is sensitive to inlet conditions, engine gimbaling could alter inlet velocity profiles and affect injector flow. If necessary, flow straighteners can be used to stabilize flow distribution, provided the pressure budget allows. Ports for pressure and temperature monitoring are generally easy to integrate, though specific

placements such as faceplate temperature sensors require early design consideration.

To verify the structural integrity of the injector, a simplified static structural analysis is a required step of the design process. Precisely determining the operating conditions of the injector, specifically the faceplate temperature, requires complex simulation and is therefore not included in this methodology, but recommended as a necessary step before testing. The simplified analysis is to confirm that the design can withstand the internal injector pressure, with no back-pressure from the chamber (representative of startup conditions) and the compressive load of the engine onto the thrust structure.

3. CFD analysis and optimization

In its first phase, this methodology focuses on analytical and empirical design for injector elements, with practical considerations for manifolding and faceplate patterns. A limitation of existing impingement studies is that they assume ideal conditions and nominal orifice flow rates, which are difficult to achieve in practice due to manifold pressure variations from inlet offsets, pressure losses, turbulence, chamber fluctuations and manufacturing tolerances.

To approximate nominal flow rates as closely as possible and achieve expected engine performance, the second phase uses CFD simulations to refine manifold flow distribution. The goal is to keep each orifice within a recommended 5 % tolerance of the nominal flow rate. The CFD optimization phase can be seen as the last few steps in the design methodology in Fig. 1.

The recommended target value of 5 % flow rate deviation is considered to be achievable for almost every orifice in the manifold, given the minimum designed pressure drop of 15 %. In some circumstances, the 5 % tolerance cannot be achieved due to certain manifolding and design decisions. In these instances, the target value can be adjusted to accommodate specific design constraints. However, the implications of this adjustment with respect to combustion performance and potential instability (although impossible to quantify without in-depth CFD), must be in aligned to the engine specifications and risk-tolerance inputs.

3.1. Models

In the penultimate step of the methodology, the CFD analysis of the injector flow domains aims to solve for the injector pressure drop and flow rate through each orifice. These data are used in the iterative CFD optimization loop to inform design changes and maximize the uniform distribution of propellant. This section details the selected numerical methods and models to be used in the CFD analysis. The analysis of an injector developed according to this methodology demonstrates the pre-processing, post-processing and interpretation of the simulation and resultant data

The flow is assumed to be steady, incompressible and turbulent in the analysis and can be suitably modeled with the Reynolds-Averaged Navier-Stokes (RANS) equations. An example of such an analysis is provided below. Here, an injector design was developed according to the methodology and then subjected to CFD simulation using the Semi-Implicit Method for Pressure Linked Equations (SIMPLE) algorithm and a second-order upwind discretization scheme.

The selection of a RANS turbulence model is based on the work done in testing, modeling and optimizing flow distribution in manifolds by Chen and Sparrow [59,60] where the application is functionally similar to the distribution in an injector manifold. The study's outcome is that the Realizable K-Epsilon turbulence model yielded the best fit to experimental data and is therefore the suggested turbulence model for the injector mass flow rate distribution analysis described here

The viscous sub-layer in the boundary layer is resolved with a low- y^+ approach to better predict losses and injector pressure drop. A wall y^+ value of less than 1.0 is targeted across the surface. However, given

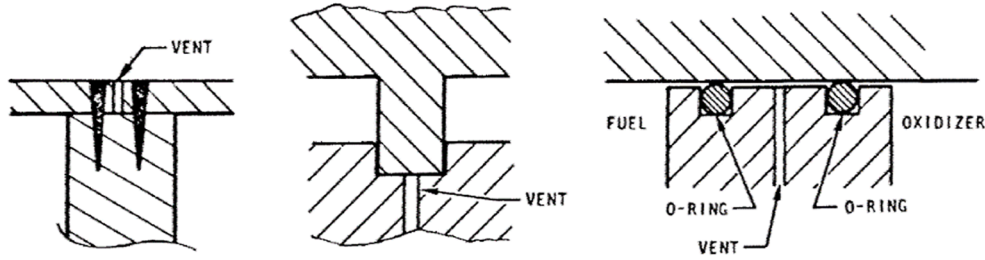


Fig. 7. Leak paths used for inter-propellant sealing [21].

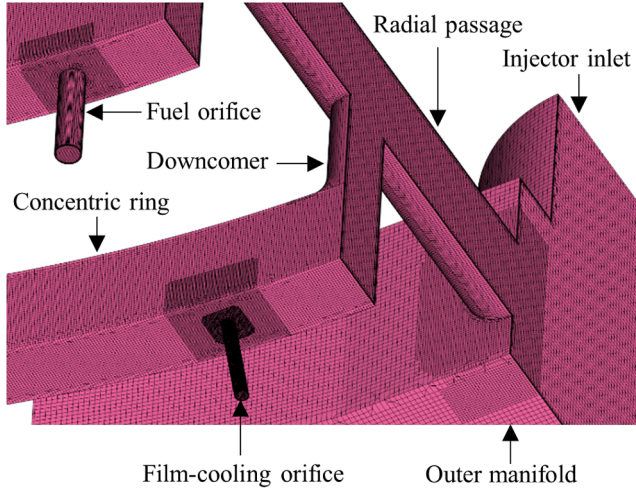


Fig. 8. Underside view of Injector A's fuel domain mesh.

the large variation in geometry and therefore velocities across the manifold, maintaining y^+ values below 1 across design iterations is time-consuming. Therefore, at least 90 % of the manifold surface, and 100 % of the orifice surfaces must meet this condition, where a blended approach is used for y^+ values out of this range.

The inlet is a uniform velocity boundary condition, calculated from the inlet area and mass flow rate. The orifice outlets are set as a pressure outlet boundary condition, where the pressure is the absolute chamber pressure [59] of 12 bar. For the present example, densities and viscosities from Table 2 were used for the simulated fluid properties.

3.2. Numerical grid

A parts-based meshing approach is used for both fluid domains. Mesh refinements are defined for each geometric feature such as the manifold, downcomers and orifices, which ensures consistent mesh quality across the different design iterations. A hexahedral type mesh, although less efficient than a polyhedral mesh, is chosen for the simplicity and predictability of controlling mesh refinements across multiple designs.

A base cell size of 0.3 mm is used for the large volume of the manifold with no refinements. The hex type mesh is then refined by one level (to 0.15 mm) for the downcomers and internal flow passages, and refined one level more (0.075 mm) for the orifices. Film-cooling orifices of the JET A-1 domain are refined yet another level further to 0.0375 mm, due to their small diameters. Fig. 8 illustrates the high-resolution mesh of the kerosene flow domain in the concentric ring injector, referred to as Injector A. Twenty prism layers are specified along the walls with a growth rate of 1.3 and overall total thickness of the surrounding base size, to capture the viscous effects of the boundary layer.

Table 4

Empirical vs simulated pressure drops.

Parameter	Units	LOX	JET A-1
Orifice diameter	mm	1.4	1.5
Empirical pressure drop	bar	3.94	2.27
Simulated pressure drop	bar	3.97	2.17
Percentage difference	%	0.76	1.40

The Grid Convergence Index (GCI) [61] is used to determine mesh independence, where a GCI value between 0.98 and 1.02 is specified. Injector inlet pressure and all individual orifice flow rates are investigated for grid convergence. A coarse, medium and fine mesh, with base sizes of 1.2 mm, 0.6 mm and 0.3 mm respectively, are used sequentially to initialize the flow domain, as well as check mesh independence for each design configuration.

The pressure drop across the orifices can be verified by comparing the results to values obtained from Eq. (1). The plane across which the average pressure upstream of the orifice is sampled is indicated in Fig. 9. Table 4 lists the calculated and simulated pressure drops for Injector A and the relative difference between them.

3.3. Distribution optimization

A sensitivity analysis of the manifold geometry is essential to identify which geometric modifications affect the orifice flow rate distribution and to what degree. Key geometrical features, such as downcomer diameters, downcomer lengths, radial passage diameters, and baffles/flow deflectors, can be optimized to achieve a desired flow distribution. This analysis is only required at the initial stage of the optimization process; however, it should be revisited if substantial modifications to the manifold design significantly alter the dominant flow behavior.

Injector pressure drop and individual orifice flow rates are the key variables of interest. Post-processing and interpreting varying flow rates through numerous individual orifices is best done methodically to make informed design decisions. Two post-processing strategies are used in this methodology, with both using the percentage difference between the solved orifice flow rate and nominal orifice flow rate, referred to as deviation.

The first post-processing strategy is best suited for the initial iterations of the CFD optimization loop and is referred to as the average ring distribution. This involves averaging the flow rate of orifices that share the same radial positioning in the manifold. Because of the circular pattern resulting from the annular patterning method in the injector design loop, orifices positioned on the same pitch circle diameter (PCD) will experience similar deviations in flow rate. Grouping these data points allows for quick and efficient identification of problem areas in the manifold and the effect of design changes on several orifices.

Fig. 10 shows the first post-processing strategy as a 3D graph for Injector A. Here the percentage flow rate deviation for each concentric ring is compared across the multiple design iterations. Because the

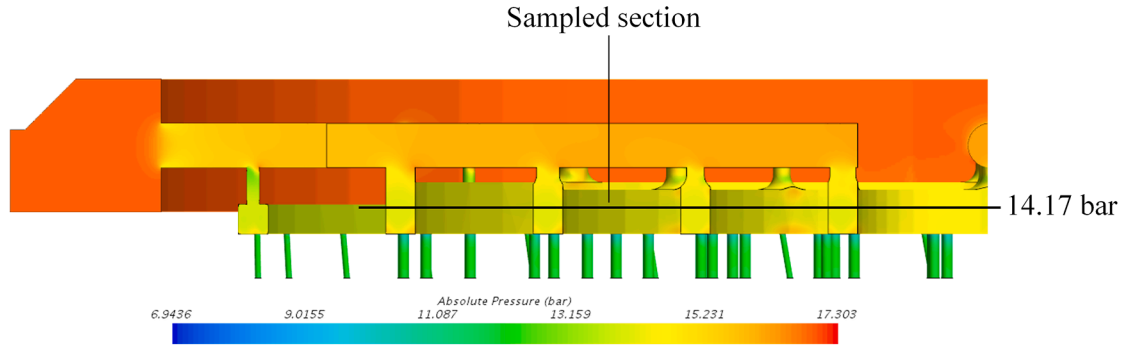


Fig. 9. Average static pressure sampled from the midpoint of the concentric ring volume.

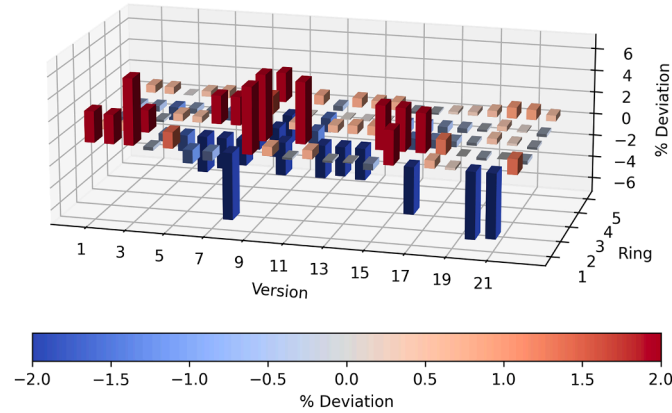


Fig. 10. Average orifice flow rate deviation of multiple design iterations, grouped by PCD.

variable being monitored is an averaged value, a tighter tolerance of 2% is applied before investigating individual orifice flow rates.

For the final iterations, the average ring distribution does not provide sufficient resolution to continue the optimization of the flow domain. The second post-processing strategy, therefore, monitors and plots individual orifice flow rate deviation as a heat map. The LOX flow deviation heat map for one of the ABLE injectors is shown in Fig. 11 as an example. The percentage deviation values for each orifice outlet are plotted at the X-Y coordinates of the orifice center, where (0 mm, 0 mm) is the center of the faceplate. The deviation is then linearly interpolated between orifices outlets.

The deviation heatmaps must not be confused with actual propellant distribution in the combustion chamber. They only represent the values at the orifice outlet before impingement and do not account for the distribution of the propellant after impingement and droplet dispersion. Nevertheless, the heatmaps serve usefully to inform design decisions based on flow rate distribution.

The process to achieve uniform mass flow distribution requires multiple iterations and highlights required design changes, such as the positioning of propellant inlets or downcomers. Once all orifice mass flow rates are within the targeted tolerance, the design of the injector flow domain can be considered finalized. In some instances, achieving perfectly uniform distribution across each individual orifice is not possible due to earlier manifold and design choices. The severity and extent of the maldistribution must then be compared to compromises associated with the design decisions made. For example, a LRLP approach that employs an integral type manifold has far fewer geometrical features to affect control over orifice flow distribution. If there are only two off-nominal orifices near the center, such as in Fig. 11, this may be considered acceptable, while a distribution that results in high concentrations of oxygen near the wall may not.

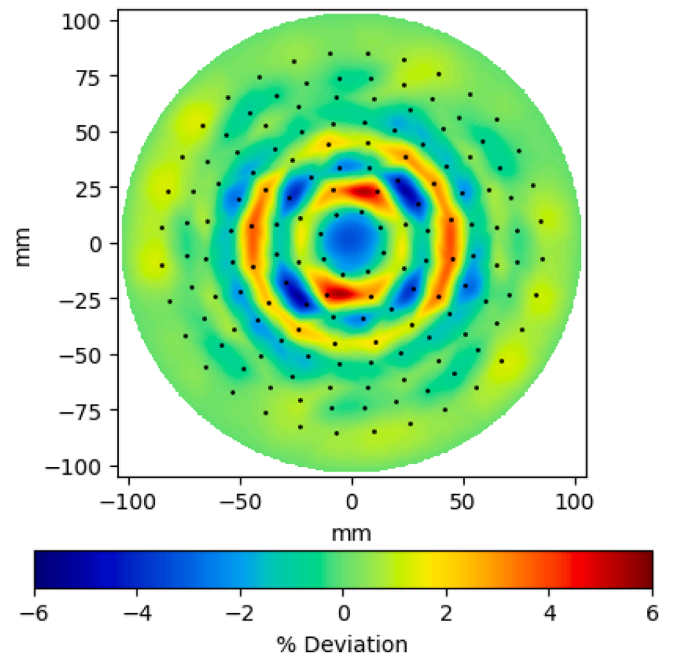


Fig. 11. Orifice flow distribution for Injector A LOX domain.

At this point, the injector design output from the methodology is complete. Further investigations, such as atomization and combustion studies on the specific element design, can be conducted if deemed

Table 5
Injector A inputs.

Input parameter	Input variable(s)
Risk-tolerance	HRHP
Engine specifications	ABLE engine
Manufacturing methods	CNC, EBW, brazing
Materials	316 Stainless steel, C10100 copper

necessary. Depending on the results of the initial structural analysis, a more comprehensive study on the structural integrity of the injector may also be required before any testing.

4. Injector designs

The two injectors are designed for the 18 kN-thrust ABLE engine. They therefore share the same engine specification input, as shown in Table 1, as well as engine interface requirements. However, each injector uses notably different risk-tolerance and manufacturing inputs for the design methodology.

The first injector design for the ABLE engine, referred to as Injector A, was designed to achieve higher performance with an increased risk of combustion instability and component failure. It therefore embodied a high-risk-high-performance (HRHP) input and a higher budget, allowing for more complex manufacturing and development. The second injector (B) utilized a more conservative approach regarding engine performance and manufacturing based on a low-risk-low-performance (LRLP) input.

In line with the SAFFIRE program's objective to develop liquid rocket engines within the South African industrial context, both injector designs were constrained to traditional manufacturing methods commonly used in the local manufacturing base. At the time of these designs, additive manufacturing was still maturing and therefore not considered. Brazing and electron beam welding (EBW) were accessible for joining dissimilar metals, specifically the injector faceplate and body. The ABLE engine was a technology demonstrator intended exclusively for hot-fire testing and not flight; therefore, engine mass was not a concern.

4.1. Injector A

The key design decisions for the first injector are listed below, based on the inputs listed in Table 5.

1. The element choice is a triplet in the O-F-O configuration.
2. A concentric ring manifold was selected to enable higher element density and an element orientation that allowed better combustion chamber wall compatibility.
3. Film-cooling was included as per the engine requirements.
4. Copper was selected as the faceplate material, following a conservative approach regarding the unknown heat flux.
5. 316L stainless steel was selected as the primary body material.
6. The copper faceplate was brazed to the stainless steel body.

The final Injector A design can be seen in Fig. 12. The LOX travels through the inlet, which is in line with the central axis of the engine, into the LOX dome. From the LOX dome, LOX goes through the downcomers and into every alternating concentric ring. JET A-1 enters at the outer perimeter of the injector, making its way to each other concentric ring via radial passages and downcomers. Fig. 13 shows the injector body and faceplate in isolation, where the flow paths to each alternating ring for LOX and JET A-1 can be seen in detail.

4.1.1. Element pattern

The number of orifices and their diameter, length and impingement distance all effect the overall footprint/total width of the element. This has a direct effect on the spacing and total element pattern, as discussed in Section 2.4. Iterating between the number of elements and orifice

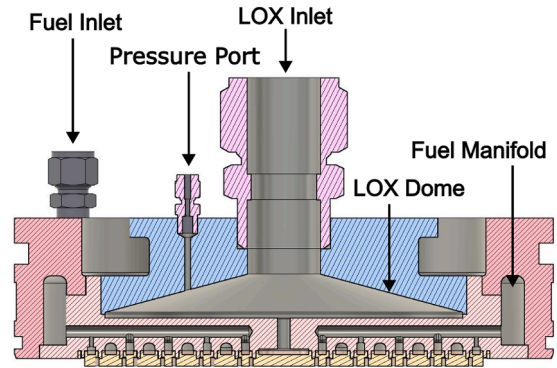


Fig. 12. Injector A final design cross-section.

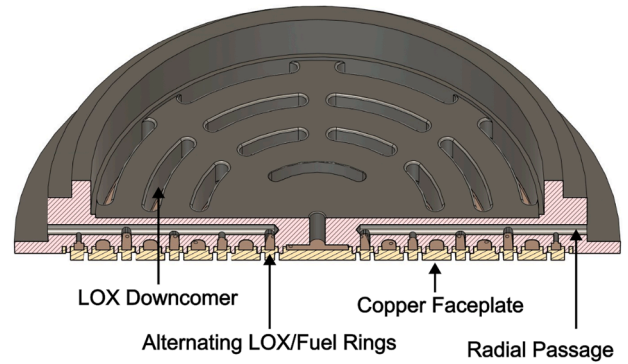


Fig. 13. Injector A body showing LOX downcomers.

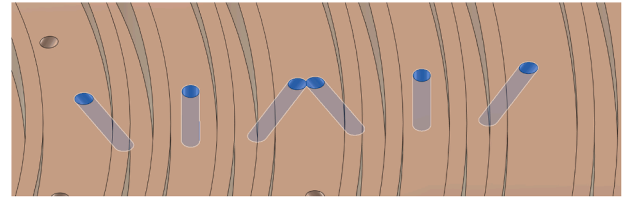


Fig. 14. Intersecting orifice inlets.

size, the width of the annular rings, G , was set to 20 mm, allowing four annular rings of the same width to fit onto the faceplate, as shown in Fig. 3. Eight elements were specified in the first concentric ring. Following the annular patterning method, a total of 80 elements were evenly distributed across the faceplate.

The triplet elements were rotated 15° about the center-line to avoid interference at oxygen orifice inlets shown in Fig. 14. Rotating the triplet elements in this manner reduces the intersection of the spray fans. Although this impacts the inter-element mixing, the triplet element has good mixing characteristics [7] and does not rely as heavily on inter-element mixing to perform well. This approach has also been used by Wheeler and Kirby [62] and Lawver et al. [63]. The angled orifices also enabled a smaller annular width, G , allowing for a denser element pattern. Table 6 details the parameters of the annular patterning method for this design. Fig. 3 shows the final element pattern for Injector A with the corresponding ring numbers, n .

4.1.2. Element design

After the element pattern and the total number of elements have been determined, the flow rate through each orifice is known and the orifice geometry can be finalized. The O-F-O triplet element design methodology detailed in Section 2.3.1 and Fig. 2 is used to determine suitable orifice combinations for appropriate faceplate thicknesses, pressure drop and diameters suitable for manufacture. As per the methodology,

Table 6
Element pattern parameters for Injector A.

Element ring (n)	Ring area (A_r)(mm ²)	Elements per ring (E_n)	Element area (A_e)(mm ²)
1	2 513	8	314.2
2	5 027	16	314.2
3	7 540	24	314.2
4	10 053	32	314.2
Faceplate	33 654	80	420.7

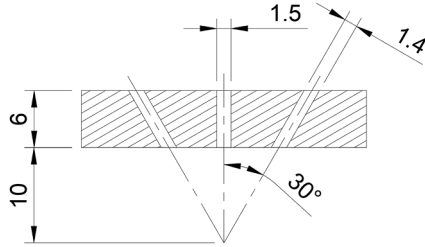


Fig. 15. Injector A triplet element.

Table 7
Element pattern parameters for Injector A.

Parameter	Unit	Value
JET A-1 orifice velocity	m/s	18.16
LOX orifice velocity	m/s	20.12
Ferrenberg value [26]	–	1.22
JET A-1 pressure drop	bar	2.26
LOX pressure drop	bar	3.93

orifice diameters were selected based on pressure drop and maximizing atomization performance. A 60° total impingement angle is selected due to its effective mixing characteristics [5,21,43]. The chosen element design is shown in Fig. 15 and associated performance parameters are detailed in Table 7.

The ABLE engine required 10 % of the JET A-1 flow rate for film-cooling. The film-cooling orifices are non-impinging, shower head type injection elements. In this design, 32 film-cooling elements were selected to match the number of injector elements near the wall. The film-cooling orifices were sized at 0.8 mm in order to have a similar pressure drop as the JET A-1 orifices (1.75 bar).

4.1.3. Manifold design

The manifold design is shaped by manufacturing and structural constraints to achieve a dense element pattern. While any element pattern could theoretically be used, the manifold type, specific design, and material considerations ultimately define feasible patterns. Injector A employs a concentric ring injector design that supports the triplet element, offering the advantage of using dissimilar materials and traditional manufacturing methods. The design uses a 316 stainless steel body to channel LOX and JET A-1 into concentric rings. A copper faceplate is brazed to the stainless steel body via a tongue-and-groove joint to maximize the brazed surface area, ensuring a robust connection through vacuum brazing with silver solder.

After brazing and joining, leak paths were cut into the faceplate to separate the solder joint between the LOX and JET A-1 domains. This post-processing step eliminates the risk of inter-propellant mixing in the manifolds due to brazing leaks, creating a direct pathway for the propellants into the combustion chamber and ensuring operational safety.

4.1.4. CFD optimization

The CFD optimization phase follows the methodology outlined in Section 3. The initial sensitivity study showed that both flow domains

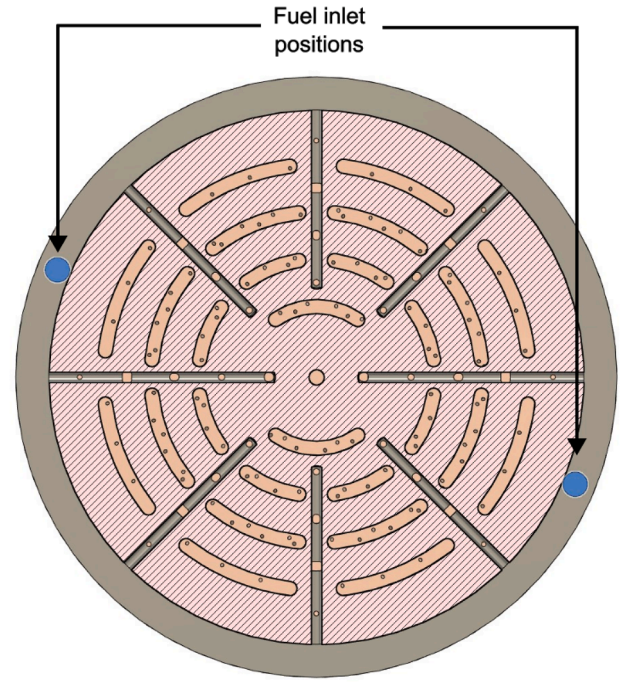


Fig. 16. Injector A manifold.

were sensitive to inlet conditions. For the LOX domain, this was addressed by setting the inlet to come down through the engine's center line. For the JET A-1 domain, two inlets were positioned on opposing sides of the manifold, as shown by the blue circles in Fig. 16. The cross-sectional top view of the stainless steel body of Injector A shows the JET A-1 radial passages between the LOX downcomers. The different shaped and sized JET A-1 downcomers can be seen in each radial passage.

The LOX and JET A-1 domains were analyzed independently to optimize flow distribution. For the JET A-1 domain, adjustments to radial passages had minimal impact, while varying the size, shape, and chamfers of the downcomers significantly influenced the distribution across concentric rings. Flow rates through orifices within the same ring were consistent except when directly below a downcomer, emphasizing the importance of angular positioning during design.

In the LOX domain, larger downcomers allowed more flexibility, but maximizing their size improved flow uniformity. Manipulating the inlet radii of the downcomers effectively influenced distribution. Initial optimization focused on minimizing aggregate flow deviation per ring, achieving less than 2 % deviation, as shown in Fig. 10, before analyzing deviations at the per-orifice level.

Final flow deviation heat maps for LOX and JET A-1 domains are presented in Figs. 11 and 17. All orifices met the 5 % deviation target, with some orifices towards the center showing relatively higher deviation. Attempts to address this local deviation, such as modifying downcomer size, inlet radius, diffuser angle, or faceplate rotation, were ineffective without compromising overall uniformity. The final design provided the most uniform distribution across the injector faceplate.

The CFD results give a total injector pressure drop (from manifold inlet to orifice outlet) of 3.85 bar for the LOX domain and 4.33 bar for the JET A-1 domain. The CFD methodology used shows good agreement in the orifice pressure drop (refer to Fig. 9 and Table 4), therefore the difference between the CFD result and expected orifice pressure drop is attributed to the pressure losses in the manifold. The total LOX pressure drop is similar to the orifice pressure drop, implying negligible manifold losses, whereas the JET A-1 domain shows manifold losses of approximately 2.07 bar.

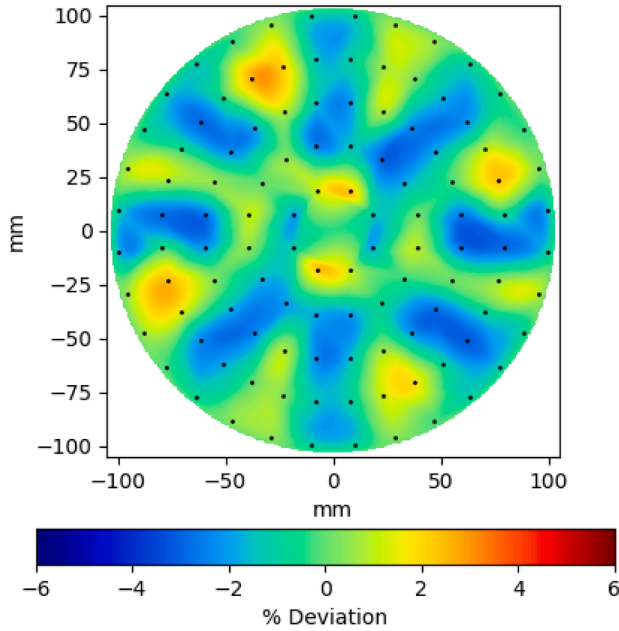


Fig. 17. Orifice flow distribution for Injector A JET A-1 domain.

Table 8
Injector B inputs.

Input parameter	Input variable(s)
Risk-tolerance	LRLP
Engine specifications	ABLE engine
Manufacturing methods	CNC, EBW
Materials	316 Stainless steel, C11000 copper

4.2. Injector B

The second injector was designed as an LRLP alternative to Injector A. This approach favors a more practical/robust design that is simpler to manufacture with fewer likely points of failure, but with the potential penalty of performance loss. The input parameters for the design methodology are detailed in Table 8, and the resulting design decisions are listed below:

1. The selected elements are like-doublets.
2. An integral type manifold was selected. Creating the injector core from a single body limits the possible configurations but drastically reduces the complexity and points of failure, reducing manufacturing cost and risk.
3. Film-cooling was included as per engine requirements.
4. Copper was selected as the faceplate material as a conservative choice. Due to the integral manifold design, this material was also used for the primary body.
5. 316L stainless steel was selected as the outer manifold material.
6. The copper body and stainless steel manifold were joined using Electron Beam Welding.

Using an identical manifold as Injector A (Fig. 12), the final design for Injector B is shown in Fig. 18. The LOX dome distributes propellant to each drilled downcomer that intersects with the like-doublet orifices. The JET A-1 manifold and radial passages supply JET A-1 to each orifice inlet.

4.2.1. Element pattern

The like-doublet injection element relies on secondary mixing, with LOX and JET A-1 injection elements alternating across the faceplate to

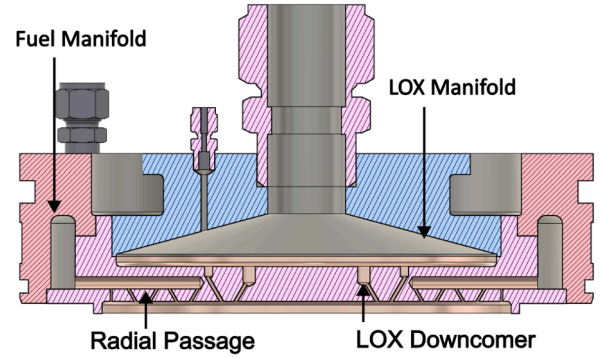


Fig. 18. Injector B final design cross-section.

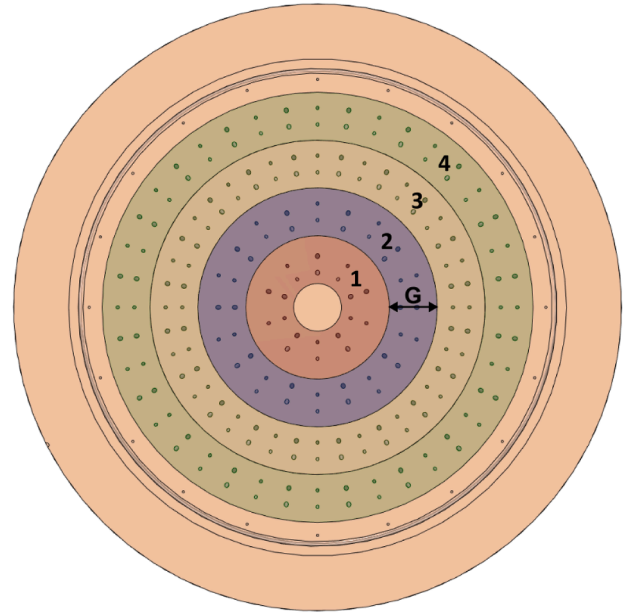


Fig. 19. Injector B faceplate element pattern.

Table 9
Element pattern parameters for Injector B.

Element ring (n)	Ring area (A_n)(mm ²)	Elements per ring (E_n)	Element area (A_e)(mm ²)
1	2 513	10	251.3
2	5 027	20	251.3
3	7 540	40	118.5
4	10 053	40	251.3
Faceplate	33 654	110	305.9

promote inter-element mixing through intersecting spray fans. The selected pattern, shown in Fig. 19, uses 20 mm annular ring widths, as in Injector A, accommodating four rings and 110 injection elements. Parameters for the layout are provided in Table 9. Due to manifold constraints, the number of elements does not increase linearly with ring number, leading to higher element density in ring 3. The layout minimizes this deviation to maximize distribution uniformity.

4.2.2. Element design

The element design of the like-doublet is simpler than that of the triplet. The strategy for doublet elements is to maximize the distribution while meeting the pressure drop requirements. The manifold design primarily constrains the distribution, where 110 elements are arranged. These injection elements are divided equally into LOX and JET A-1 elements, where the JET A-1 elements are notably smaller to maintain

Table 10
Element pattern parameters for Injector B.

Parameter	Unit	Value
JET A-1 orifice velocity	m/s	17.58
LOX orifice velocity	m/s	17.71
JET A-1 pressure drop	bar	2.12
LOX pressure drop	bar	3.04

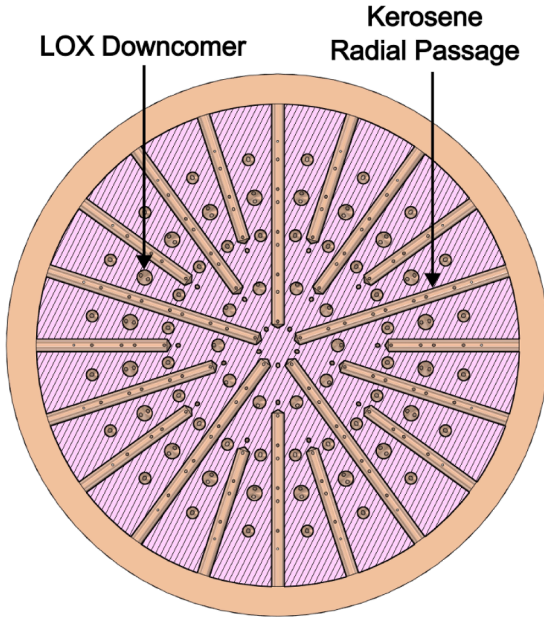


Fig. 20. Injector B cross-section of the integral manifold body.

the same pressure drop for a lower flow rate. The element performance parameters are detailed in Table 10.

Unlike the concentric ring injector, which has an entire annular ring for film-cooling orifices, the integral injector can only locate film-cooling orifices beneath radial passages. This severely limits the number of orifices and is a notable disadvantage of an integral style injector in this configuration. Therefore, 20 film-cooling orifices with a diameter of 0.95 mm, giving an estimated pressure drop of 2.25 bar.

4.2.3. Manifold design

The manifold design of an integral injector reduces manufacturing complexity but limits element patterns. Still, it enables a more robust design by eliminating potential failure modes such as leakage through seals and brazing. Like doublets were selected for Injector B, which requires intersecting spray fans for sufficient mixing. Therefore, LOX and JET A-1 elements were orientated in the same direction so that when placed next to each other, the spray fans would intersect. A cross-section showing the alternating JET A-1 radial passages and LOX downcomers can be seen in Fig. 20.

The alternating pattern of radial passages used five passages as a base, from which the rest of the pattern was derived. This base pattern maximized element distribution for this specific chamber. The diameter of the radial passage was limited by the high aspect ratio of the drill bit required for this machining operation. LOX downcomers were drilled at varying depths between the radial passages to intersect the orifice inlets, as shown in Fig. 18.

The faceplate thickness was determined by the minimum L/d ratio of 4 for the JET A-1 orifices, along with structural simulations. Cold rolled copper plate was selected for the body material due to the additional strength gained through work hardening even though, for such a thick plate, the hardening effect was not expected to

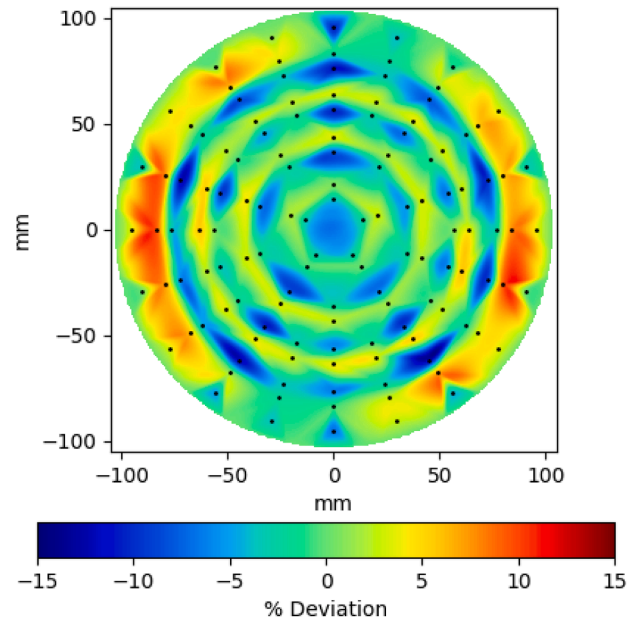


Fig. 21. Orifice flow distribution for Injector B JET A-1 domain.

penetrate through the entire depth of the raw stock. The structural simulations, therefore, used conservative material properties while, in practice, the active cooling of the copper by the internal flow of cryogenic oxygen was expected to strengthen the injector during operation.

4.2.4. CFD optimization

The same structured CFD analysis and optimization approach was applied to Injector B. The simpler manifold geometry reduced the number of adjustable parameters, particularly in the JET A-1 domain, resulting in fewer design iterations compared to Injector A. Flow distribution in the JET A-1 domain was most sensitive to the angular position of the manifold inlets. Increasing the diameter of radial passages was not feasible as it risked intersecting LOX orifices (Fig. 20), while reducing the diameter incrementally added manufacturing complexity without significant improvement. The orifice angle caused a bias toward inward-facing orifices, as clearly seen in the final JET A-1 flow distribution results (Fig. 21). Although the total flow deviation per orifice was higher than Injector A, this trade off was acceptable in favor of a simpler and more robust design.

In the LOX domain, flow distribution was most influenced by the downcomer diameter. Oxidizer orifice inlets near the center of the injector, positioned below the manifold inlet, experienced higher flow rates, which were mitigated by reducing the downcomer diameter in this region. The longer orifices required for intersecting the downcomers had minimal impact on pressure loss. The final LOX flow distribution results are shown in Fig. 22.

The CFD results give a total injector pressure drop (from manifold inlet to orifice outlet) of 2.76 bar for the LOX domain and 2.93 bar for the JET A-1 domain. The manifold losses follow the same trend as Injector A, where the LOX manifold losses are negligible and the JET A-1 domain shows a manifold loss of 0.81 bar.

5. Cold-flow testing

Cold-flow testing is conducted using water as an analog fluid to evaluate flow distribution performance, identify potential manufacturing or design defects and thereby assess the effectiveness of the design methodology. Mass flow rates through individual orifices are measured and compared to CFD simulations. One method to achieve comparability between the experimental and simulated results is to apply the Reynolds

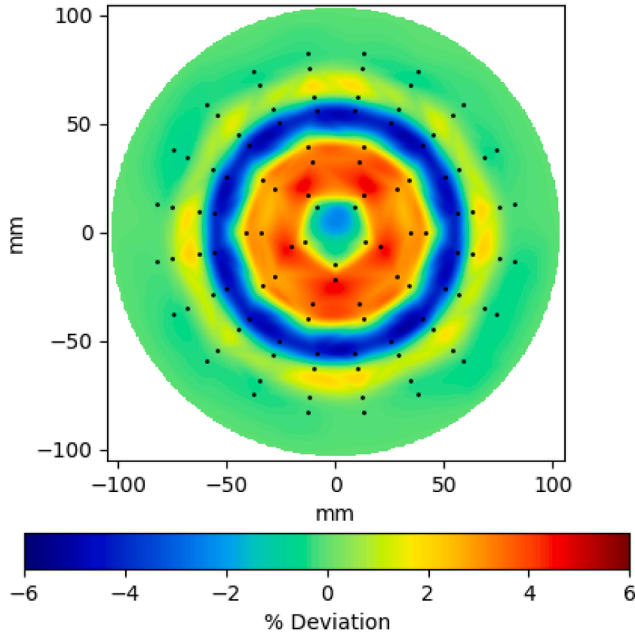


Fig. 22. Orifice flow distribution for Injector B LOX domain.

law of similarity and adjust the water flow rate to match the Reynolds number of the actual propellants. This adjustment accounts for density and viscosity differences, requiring a flow rate reduction of approximately 0.5 for JET A-1 and 4.5 for LOX. However, this method does not account for potential cavitation during testing at atmospheric conditions.

Nominal engine operation involves chamber back-pressure, ensuring that propellant pressures remain above vapor pressure to avoid cavitation. Cold-flow testing without a pressure chamber increases the risk of cavitation, which affects pressure drop and flow distribution [64]. To mitigate this, water flow rates are reduced to prevent cavitation, determined analytically and verified through CFD simulations. While a pressurized chamber could replicate nominal back-pressure, its absence necessitates adjustments to the analog mass flow rates, deviating from nominal Reynolds numbers. In this study, simulations at reduced flow rates were compared with experimental results, and any discrepancies served as indicators of potential deviations under nominal operating conditions.

5.1. Methodology

A test stand was constructed for the water cold-flow tests. This comprised a 380-liter pressure vessel, an aluminum frame housing the injector and orifice collection rig, a regulator for pressurant supply, control valves and pressure transducers. Static pressure transducers were located at the nitrogen pressurant cylinder, pressure vessel, supply line and injector manifold, as shown in Fig. 23. The flow rate was controlled by adjusting the nitrogen supply pressure.

The flow rate was measured by time-averaging the total mass of water collected over the test duration. The relatively constant injector pressure resulted in a close-to-constant flow rate. The start-up and shutdown transience were visible in the pressure data. The effect of this transience on the overall mass collected was investigated by specifically capturing the mass flow rate over the transient period. It was found that the transient period was sufficiently short that the mass flow of both the start-up and shutdown process accounted for less than 0.03 % of the total mass for long-duration tests (greater than 10 s). The effect of transience was therefore considered insignificant and disregarded.

The water cold-flow tests were conducted at set supply pressures to determine the flow coefficient through the injector [65]. The flow coefficient,

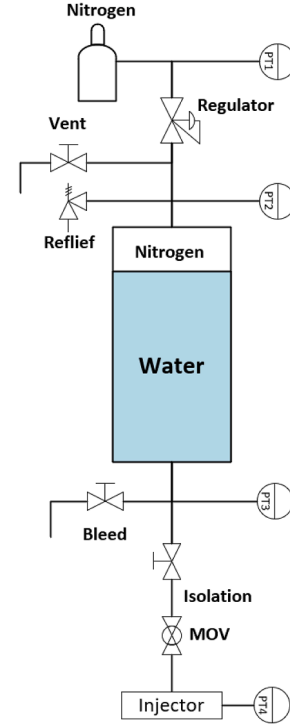


Fig. 23. Cold-flow test stand P&ID.

as determined by Eq. (8), was used to compare data at similar flow rates. The sensitivity of the mass flow rate to supply pressure resulted in minor variations between each test due to the limited resolution of the manual pressure regulator used for testing. Each injector domain was tested separately.

$$K_v = Q \sqrt{\frac{SG}{\delta P}} \quad (8)$$

For distribution tests, individual orifice flow rates were measured by capturing the efflux from each orifice in a separate container. Fig. 24 shows the equipment used to isolate, distribute, and contain the water emanating from each orifice. As with the total mass flow rate measurement, the average orifice flow rate was measured and mapped by weighing the mass of each container and averaging the total orifice mass collected over the duration of the test.

5.2. Results and discussion

Each propellant domain of Injectors A and B was cold-flow tested with water at reduced flow rates to avoid cavitation within each flow domain. Pressure drops, mass flow rate, and resultant flow coefficients, K_v , were captured and compared with CFD results. The difference between the manifold pressure and atmospheric pressure was used to determine the pressure drop across each flow domain. As with mass flow rate, pressure drop was averaged over the test duration. The flow distribution associated with each injector was also determined and compared to the simulations using a heat map based on the flow rate deviation per orifice from the nominal mass flow rate.

5.2.1. Injector A

The flow rate, pressure drop and corresponding flow coefficient for both LOX and JET A-1 domains for Injector A are detailed in Table 11. The difference between the experimental and simulated flow coefficient results are 0.4 % and 5.3 % for the LOX and JET A-1 domain respectively.

The injector manifold pressures for the final tests are shown in Fig. 25. As more tests were conducted, the test duration was iteratively shortened to accommodate more efficient use of the water tank while



Fig. 24. Water cold-flow testing of Injector A JET A-1 domain.

Table 11

Averaged experimental and simulated flow parameters for Injector A.

Scenario	Flow rate (kg/s)	Pressure drop (bar)	Flow coefficient
LOX experiment	2.36	0.81	9.49
LOX simulation	2.35	0.8	9.45
JET A-1 experiment	1.04	0.63	4.73
JET A-1 simulation	1	0.64	4.49

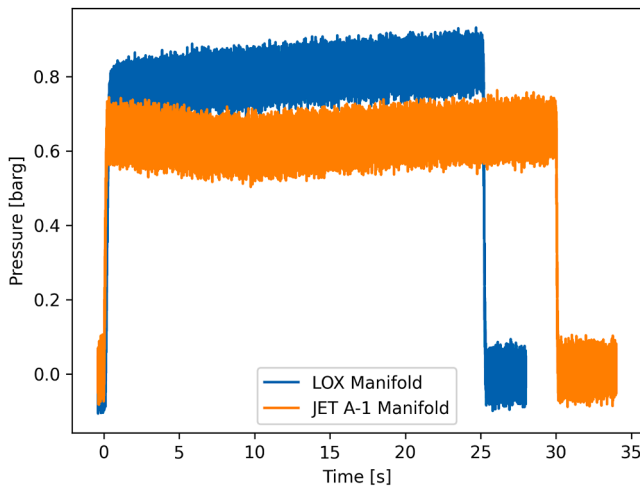


Fig. 25. Injector A manifold pressures for water cold-flow tests.

maintaining sufficient time to reach steady-state and ensure reliable data acquisition.

The flow distribution heat map for Injector A's LOX domain is shown in Fig. 26. The distribution is relatively even, except for a few outlying orifices. These variations do not follow any specific pattern and are therefore believed to be caused by poor manufacturing tolerance or defects such as burrs.

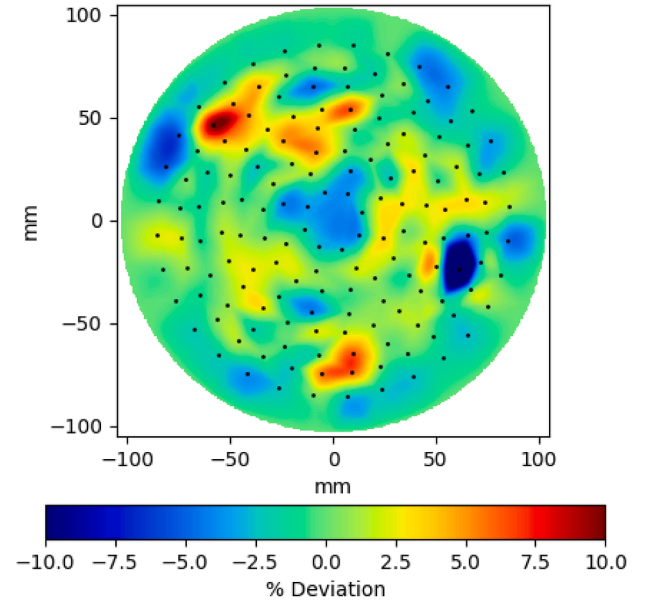


Fig. 26. Injector A LOX domain flow distribution results.

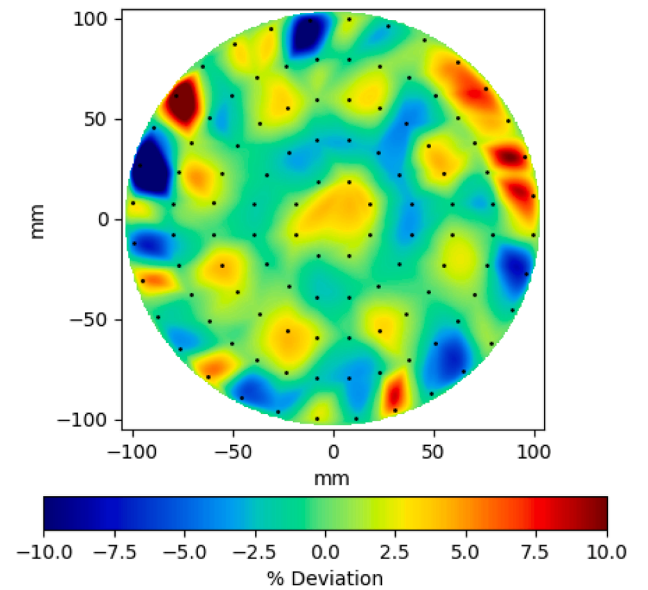


Fig. 27. Injector A JET A-1 domain flow distribution results.

The per-orifice mass flow rate data measured for the JET A-1 domain also exhibited a relatively even flow distribution. Fig. 27 shows that most orifices are within 5% of the targeted design condition. However, the outermost orifices have a higher deviation from the nominal mass flow rate. These are film-cooling orifices with a significantly smaller diameter and flow rate, and dimensional variations due to manufacturing tolerances significantly affect the cross-sectional area and, therefore, mass flow rate associated with a given orifice. A lower nominal per-orifice mass flow rate also means that slight variations in the flow rate constitute a relatively high fraction of the nominal flow. For example, the design mass flow through the film-cooling orifice is 7 g/s, where a deviation of 10% is 0.7 g/s. Compared to the 26 g/s of the primary elements, the deviation in film-cooling was not considered to be of serious concern.

Table 12
Averaged experimental and simulated flow parameters for Injector B.

Scenario	Flow rate (kg/s)	Pressure drop (bar)	Flow coefficient
LOX experiment	3.23	1.05	11.68
LOX simulation	3.29	1.1	11.32
JET A-1 experiment	1.09	0.39	6.3
JET A-1 simulation	1.1	0.48	5.7

5.2.2. Injector B

The same comparison between experimental and simulated results at reduced flow rates can be seen in Table 12. The experimental flow coefficient varies from the numerical solution by 3.2% for the LOX domain and 8.8% for the JET A-1 domain. The lower pressure drop for the same flow rate implies that the orifices are likely larger than anticipated. The variation in pressure drop of the LOX domain is within experimental error, while variation of the JET A-1 domain is attributed to poor quality control and tolerances of the orifices observed during inspection of the primary injector body.

The injector manifold pressures for the final tests of Injector B are shown in Fig. 28. The test duration for later tests was shortened as it had little effect on the quality of the data and enabled more efficient testing. Here the JET A-1 manifold was one of the first configurations tested, and the LOX manifold was one of the last configurations on the bench.

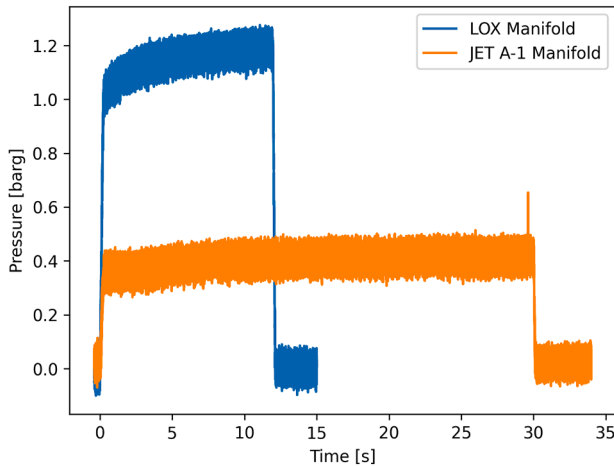


Fig. 28. Injector B manifold pressures for water cold-flow tests.

The experimental flow distribution data for the LOX domain of Injector B, as seen in Fig. 29 showed much more variation than the simulation. Injector B's LOX domain does not contain any complex internal geometries; therefore, the distribution is predominantly controlled by the diameter and quality of the orifices. For this reason, the poor distribution performance was attributed to burrs and insufficient dimensional control of the orifices, despite attempts to clean up the orifices after machining.

A heat map of the experimental data for the JET A-1 domain of injector B is given in Fig. 30 and shows more maldistribution when compared to the simulation. The maximum deviation from the nominal mass flow rate was approximately 20%. This large deviation is not unexpected based on the varying quality of the orifice inlets and tolerances.

The majority of maldistribution in the water cold-flow tests was attributed to poor manufacturing control. Injector A displayed a much more even distribution than Injector B. This can be attributed to the brazed concentric ring type manifold, which allows for deburring and polishing the orifice inlets before brazing. These activities could not be performed for Injector B due to the integral nature its manifold, where

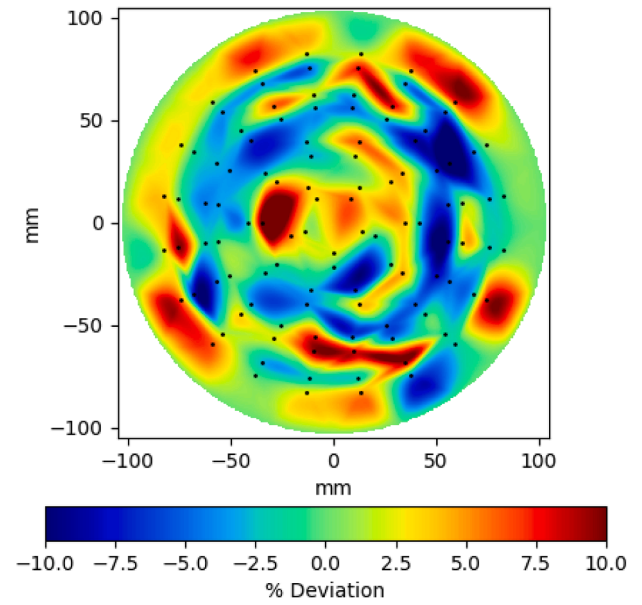


Fig. 29. Injector B LOX domain flow distribution results.

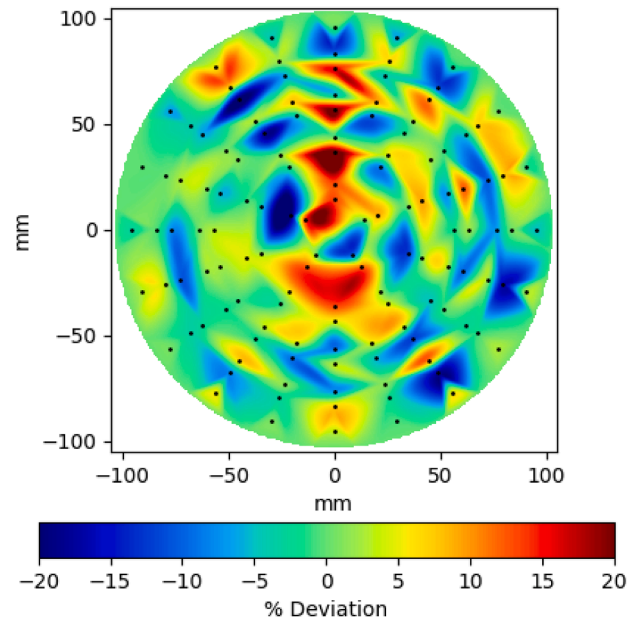


Fig. 30. Injector B JET A-1 domain flow distribution results.

orifice inlets are not easily accessible and post-processing of drilled orifices is limited. This is a known disadvantage of the integral type manifold and was considered in the design phase. Furthermore, both injectors were manufactured by different suppliers, with Injector A exhibiting appreciably better quality with respect to both the orifices and machined surfaces

Frequency analysis of the pressure transducers located in the injector manifolds provides insight into pressure fluctuations and potential flow instabilities. The transducers were sampled at 1000 Hz, which constrains the maximum resolvable frequency but some notable frequency-domain observations are still made. The Power Spectral Density (PSD) was estimated using the Fast Fourier Transform (FFT) to characterize the frequency content of the pressure signals. Fig. 31 presents the PSD of the LOX manifold pressure transducer for both injectors, while Fig. 32 shows the corresponding PSD for the JET A-1 manifold.

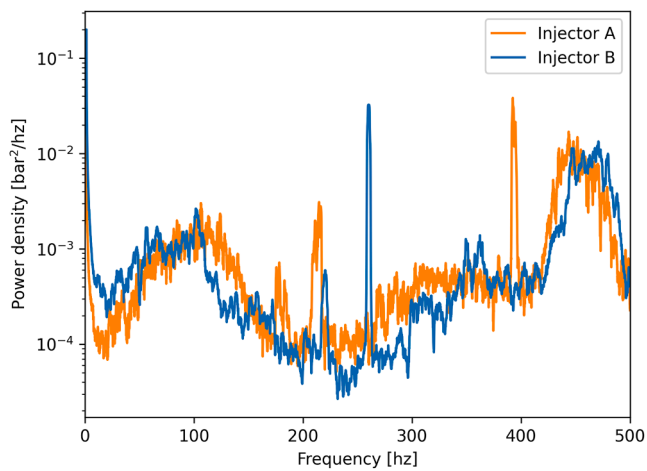


Fig. 31. PSD of LOX manifold cold-flow pressure signals.

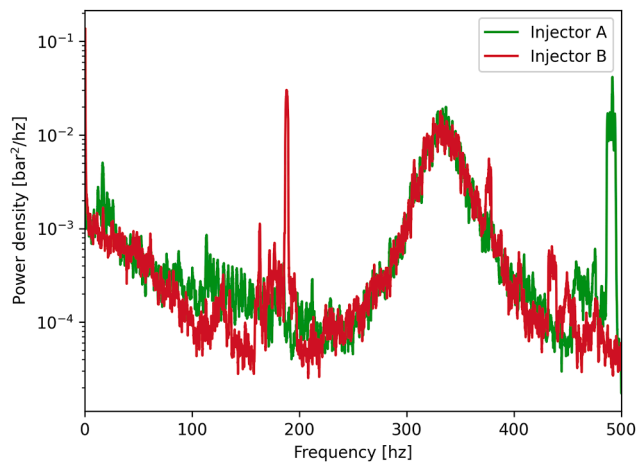


Fig. 32. PSD of JET A-1 manifold cold-flow pressure signals.

Clear similarities in the dominant frequency content of both propellant manifolds can be seen in the Figs. 31 and 32. These similarities are primarily attributed to the shared manifold geometry and experimental setup. Deviations from this trend is likely a result of the different injector configurations. In the LOX manifold, both injectors exhibit two broadband pressure oscillations in the range of 40–140 Hz and 430–480 Hz range, with distinct narrow bands at 393 Hz for Injector A and 260 Hz for Injector B. For the JET A-1 manifold, a common broadband pressure oscillation with a peak of 334 Hz is observed, accompanied by narrowbands at 490 Hz and 188 Hz for Injector A and B, respectively.

6. Hot-fire testing

Under hot-fire testing conditions, the 12 bar back-pressure of the combustion chamber ensured that the static pressure of the entire system was above both propellants' vapor pressures. In contrast to cold-flow testing, the nominal system and injector performance was expected under these conditions. In the methodology proposed here, the primary objective of hot-fire tests related to injector design is to measure pressure drop, combustion chamber stability and engine performance and compare to the designed parameters.

A leak was discovered in the film-cooling ring of Injector A was discovered during cold-flow testing. This is a result of misclamping on the copper faceplate during post-processing, breaking the seal of the brazed joint. Due to the risks associated with testing a leaking injector, and the limited time on the test range, only Injector B was hot-fire tested. The engine was hot-fire tested multiple times [29,66,67], where the hot-fire

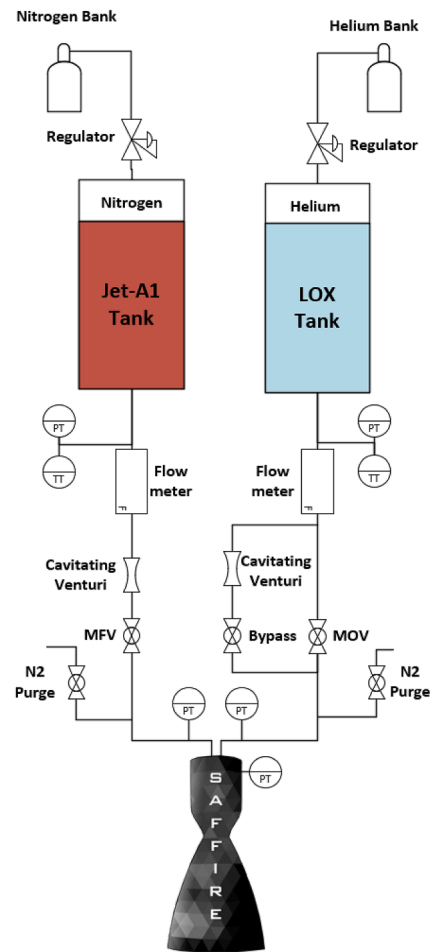


Fig. 33. Simplified P&ID of the rocket engine test facility.

test described in this section details data relevant to injector performance.

6.1. Methodology

A containerized mobile test facility was designed and constructed to test the ABE engine. This is a blowdown test facility where the liquid oxygen supply tank is pressurized by helium, and the JET A-1 supply tank is pressurized by nitrogen.

Dome-loaded regulators were used to control the supply pressure of the nitrogen and helium pressurants. These were not digitally controlled and therefore had to be set prior to the test. One of the challenges with this configuration is that on engine start-up, when there is no back-pressure from the combustion chamber, the ΔP across the system is considerably larger than that during steady operation, resulting in an above-nominal mass flow rate. This increases risk as the higher mass flow rate can result in an overpressure of the chamber and catastrophic failure of the engine on start-up.

To mitigate this risk, cavitating venturis can be used to limit the mass flow on start-up. The mass flow rate then becomes a function of supply pressure, venturi orifice diameter and fluid vapor pressure, instead of the ΔP across the system. All of these variables can be kept relatively constant to better control the propellant mass flow rates [68,69]. Fig. 33 shows a simplified P&ID of the liquid rocket engine test facility, where cavitating venturis can be seen on the JET A-1 propellant line, and the LOX bypass line.

The cryogenic pressure vessel used for LOX supply in the hot-fire testing program has a maximum allowable working pressure of 30 bar,

Table 13
Event sequence for 66 s hot-fire test.

Time	Event
T-9.5	LOX and JET A-1 purge open
T-7.5	JET A-1 purge close
T-5.0	Igniters start
T-3.5	LOX purge close
T-0.3	LOX bypass valve open
T-0	JET A-1 main valve open
T+6	LOX bypass valve close
	LOX main valve open
T+66	JET A-1 main valve close
	JET A-1 purge valve open
T+68.5	LOX main valve close
	LOX purge valve open

requiring a supply pressure well above this limit for a cavitating venturi to function reliably on the LOX propellant line. Therefore, a bypass line with a cavitating venturi at a reduced flow rate was implemented specifically for engine start-up to limit the LOX mass flow rate in the absence of chamber pressure.

Vortex flow rate meters were used on each main propellant line downstream of the tank, shown in Fig. 33, to provide a volumetric flow rate for each propellant. JET A-1 density was measured with a hydrometer prior to testing. LOX temperature was measured at the vortex flow meter, with LOX density being calculated using this temperature and a corresponding pressure measurement [70]. The propellant densities, along with the volumetric flow rate, were used to determine the mass flow rate.

Each injector had pressure transducer ports in both the LOX and JET A-1 manifolds. The pressure transducers in the LOX manifolds used a pigtail capillary tube to protect against cryogenic temperatures. The type of transducer, combined with the resonance of the capillary tubes, means that frequency data could not be used to reliably diagnose potential instabilities or resonances in the flow or equipment.

A series of propellant cold-flow tests were conducted to validate test-facility operation, system pressure drops and startup timings [71]. Purge sequences and pressures were also determined to ensure safe engine shutdown. The hot-fire test sequence was informed by these system cold-flow test data, after which an initial run of hot-fire testing was conducted. Following the successful testing of all start-up and shutdown procedures [66], a full-duration 66 s burn was conducted to investigate injector and engine performance. The start-up and shutdown sequence for this test is detailed in Table 13.

The engine was started with a reduced LOX flow rate through the bypass line fitted with a cavitating venturi. This controlled the mass flow rate on start-up and limited the pressure in the injector. Once the rate of change of pressure in the injector and chamber decreased after the chilling of the LOX line and injector, the main LOX valve was opened to achieve nominal propellant flow rates and chamber pressure. This process mitigated the risk of a hard start and associated spike in chamber pressure.

6.2. Results and discussion

The full duration burn consisted of a 6 s reduced flow, throttled combustion until the LOX line was sufficiently chilled and chamber pressure began to stabilize. This was followed by 60 s of nominal propellant mass flow rates. Fig. 34 shows the engine in operation during the 66 s burn. The associated injector and chamber pressures are shown in Fig. 35. As indicated, the LOX manifold pressure remains constant throughout the burn. In contrast, chamber pressure and JET A-1 manifold pressure fall slightly with time, likely due to excessive throat erosion of the ablative liner and an associated increase in throat diameter.

The pressure drop through the manifold remained constant because the JET A-1 mass flow was constrained by the cavitating venturi. Fig. 36



Fig. 34. ABLE hot-fire test.

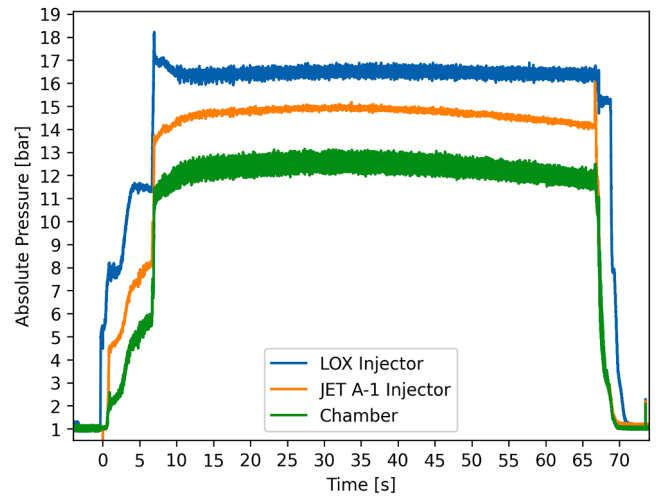


Fig. 35. Engine pressures during the 66 s hot-fire test.

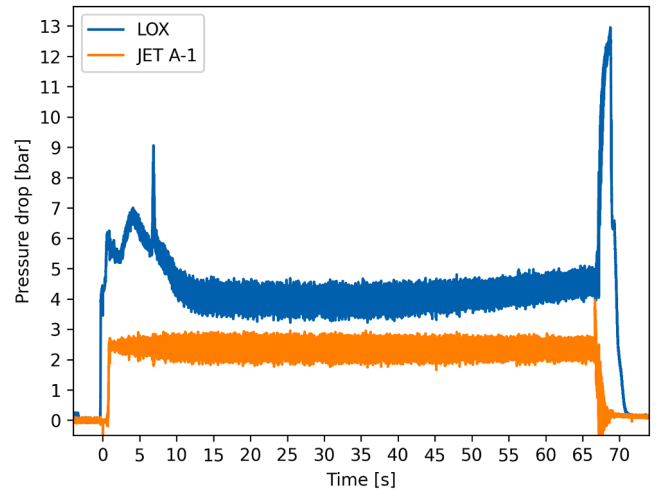


Fig. 36. Injector pressure drop relative to combustion chamber.

shows the pressure difference between each injector manifold and chamber pressure during the test. The constant mass flow rate of fuel results in a consistent pressure drop through the JET A-1 manifold. As a result, the JET A-1 manifold pressure shown in Fig. 35 is shown to be dependent on chamber pressure.

The LOX manifold pressure was not constant. Instead, the pressure difference across the LOX feed system increased as the chamber pressure decreased. Fig. 37 shows the propellant mass flow rates for the test, with the LOX mass flow rate shown to rise gradually from 5.78 kg/s at 13 s

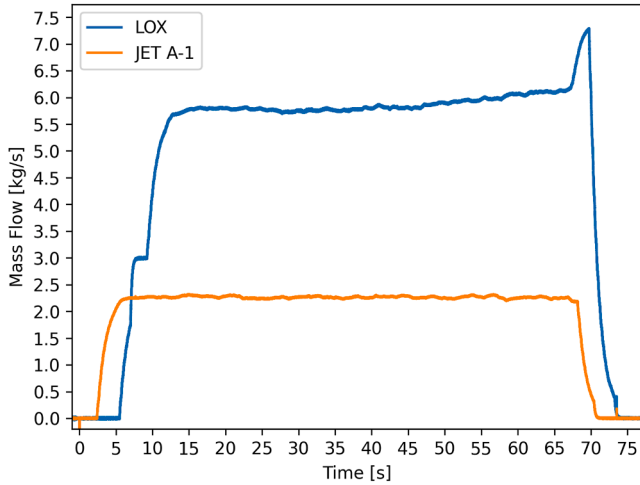


Fig. 37. Propellant mass flow rates during the 66 s hot-fire test.

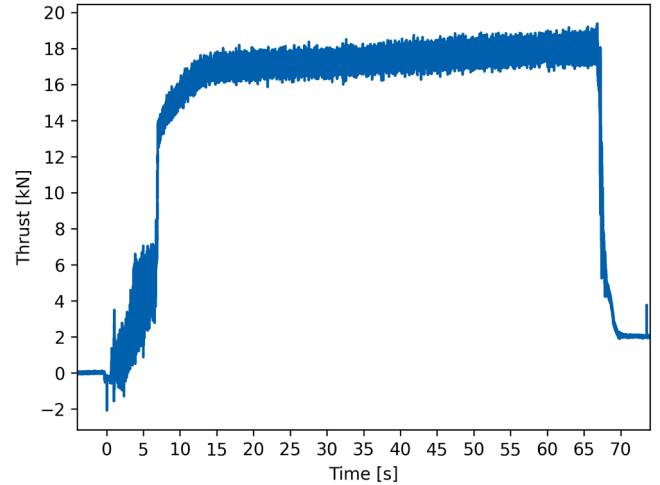


Fig. 39. Engine thrust during the 66 s hot-fire test.

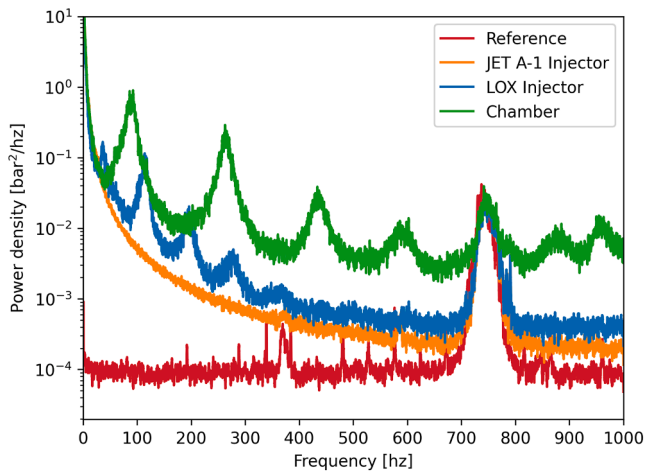


Fig. 38. PSD of injector and combustion chamber pressure signals.

into the test to 6.13 kg/s just before a sudden increase at the end of the test resulting from the engine shutdown process.

The pressure data for the hot-fire test was sampled at 2000 Hz. This relatively low frequency, combined with the capillary tubes used to thermally isolate the pressure transducers, limits the quality of the frequency data and the deductions that can be made. Reference data for a static reading of the same pressure transducer in the same experimental setup is plotted to identify which frequencies are a result of pressure oscillations and which are a result of noise. Evidently the band at 750 Hz is consistent with every pressure signal and is therefore considered noise (Fig. 38).

The JET A-1 manifold exhibits minimal resonance within the sampled frequency range, whereas the LOX manifold displays low-frequency oscillations. The relative stability observed in the kerosene manifold is likely due to the constant mass flow rate enforced by the venturi. In contrast, the combustion chamber pressure signal contains a broader range of dominant oscillations, as expected given the unsteady nature of the combustion process. Since the LOX mass flow rate is regulated by the system pressure drop (ΔP), which depends on the combustion chamber pressure, the LOX manifold is inherently more sensitive to fluctuations within the chamber. However, it remains unclear whether the pressure oscillations in the LOX manifold are a driving factor for, or a response to, the combustion chamber dynamics.

While some coupling between the LOX injector pressure and combustion chamber pressure may be present, the amplitude of the result-

Table 14

Averaged sensor data between 15 s and 20 s.

Sensor	Units	Output
JET A-1 mass flow	kg/s	2.28
JET A-1 injector pressure	bara	14.76
LOX mass flow	kg/s	5.79
LOX injector pressure	bara	16.39
Chamber pressure	bara	12.36
Thrust	kN	17.01

ing oscillations is sufficiently small to consider the combustion stable. Chamber pressure fluctuations were quantified by evaluating the peak-to-peak amplitude over 0.1 s sampling windows and comparing them to the corresponding mean pressure for each window across the dataset. The maximum observed amplitude was 5 % of the mean pressure, well within the widely accepted 10 % stability threshold [1,72].

The thrust data recorded over the duration of the test was likely affected by the expansion of both LOX and JET A-1 flexible hoses under pressure, followed by the thermal contraction of the LOX hose as it cooled to cryogenic temperatures, see Fig. 39. The changes in chamber pressure and thrust, thought to be due to the combined effect of nozzle throat erosion and the thermal contraction alluded to above, lend some uncertainty to the analysis of engine performance over time. Therefore, a more accurate assessment of the engine and injector performance is expected to be obtained during earlier stages of the test, and the final performance figures are thus determined between 15 s and 20 s from start-up. At this time, data from both flow meters had settled sufficiently following start-up. Accordingly, the averaged sensor data between 15 s and 20 s are shown in Table 14.

The chamber pressure measured is higher than the design pressure of 12 bar, and the total mass flow rate is measured to be slightly above the design point. In theory, this could either be due to the manufactured throat area being smaller than the design value or using a conservative combustion efficiency estimate during engine design. In this case, however, the throat diameter was measured to be within tolerance prior to engine testing, so the higher-than-predicted performance of the engine is attributed to two conservative assumptions. First, a c^* efficiency of 92 % was used when determining the required mass flow rate to meet the chamber pressure requirement. Second, it was assumed that 10 % of additional JET A-1 for film-cooling did not contribute to the engine's performance.

Correcting the second assumption and assuming all the injected propellant fully combusts within the combustion chamber, both the maximum theoretical and experimental c^* value are derived from the test results. The accuracy of all relevant measurements are accounted for

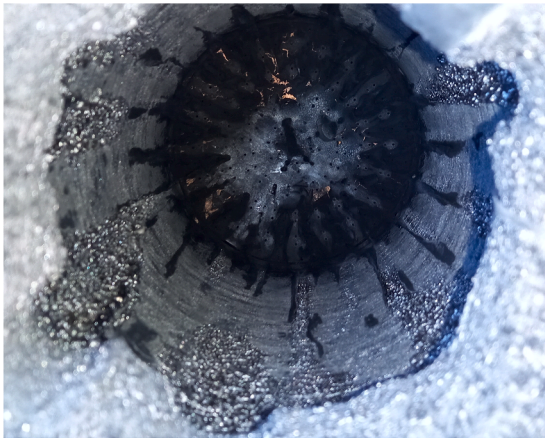


Fig. 40. Eroded ablative liner.

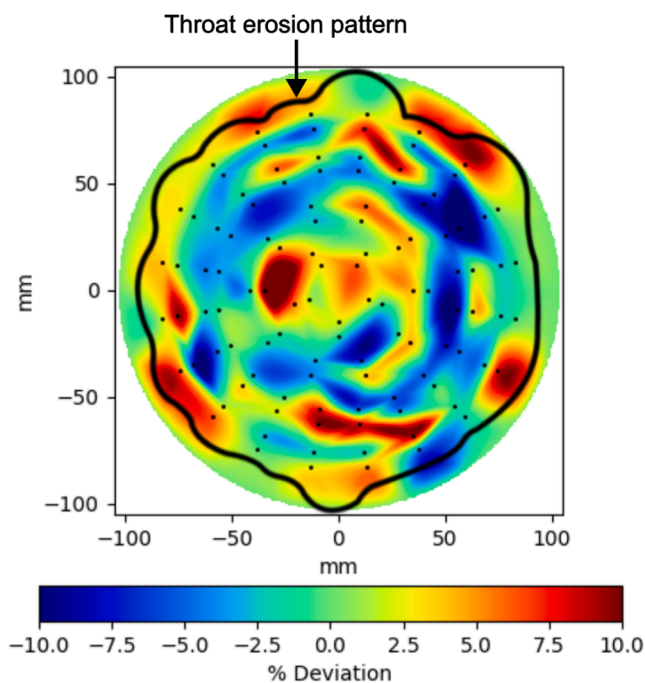


Fig. 41. LOX distribution with overlaid throat erosion pattern.

through error propagation analysis. Specifically; the error of the pressure transducer in the chamber, temperature and pressure transducers used to calculate LOX density, hydrometer for JET A-1 density, flow meters, and throat diameter measurement. The theoretical c^* for the measured O/F ratio and chamber pressure is $1758 \text{ m/s} \pm 31.5 \text{ m/s}$ and the actual value calculated from the results is $1702 \text{ m/s} \pm 30.5 \text{ m/s}$. This results in a c^* combustion efficiency of $96.8\% \pm 2.53\%$.

It is concluded that the injector and combustion chamber outperformed the slightly conservative design point, and achieved near-complete combustion. This implies that the injector performed well with respect to atomization and mixing and that the combustion chamber volume is appropriately sized.

The total injector pressure drop for JET A-1 was 2.4 bar, compared to the CFD result of 2.93 bar. This is lower than the design point, but consistent with the cold flow test results in Section 5. The LOX total pressure drop was much higher at 4.03 bar instead of the expected 2.76 bar, while the cold flow results were in agreement with the simulation. The design point used a LOX density of 1142 kg/m^3 , but in practice, the LOX tank was held at 3 bar, allowing the LOX to reach a high temperature and corresponding density of approximately 1092 kg/m^3 . Furthermore, the

temperature and pressures measured in the propellant line correspond to a density of 1032 kg/m^3 . This change in density, combined with some degree of phase change in the lines due to insufficient chill-down, are the primary explanation for the increased pressure drop through the injector.

Following the long-duration burn, significant throat erosion was observed. This level of erosion was inconsistent with other ablative liners tested during the campaign. It was noted that the eroded ablative liner was relatively soft and may have been insufficiently cured. An interesting outcome of this defect was a distinct erosion pattern at the throat, see Fig. 40.

The orientation of the injector relative to the combustion chamber was marked before disassembly to enable a comparison of the erosion pattern and injector distribution results. Fig. 41 shows the LOX distribution results from the water cold-flow tests, with the throat erosion pattern from Fig. 40 enlarged and mapped at the same angular orientation.

The large variation of the orifice flow rate in the like-doublet pairs would have resulted in many elements with a high momentum ratio between jets. This, in turn, skews the spray fan. Because the like-doublet relies on intersecting spray-fans to mix the propellants, it is highly probable that insufficient mixing occurred at the top of the combustion chamber. The erosion grooves begin just downstream and slightly off-axis of the film-cooling orifices. Accordingly, it may further be hypothesized that misaligned LOX and JET A-1 spray fans resulted in regions with excess oxygen near the wall. This excess oxygen would combust with the axial film-cooling jets and initiate the wall erosion. The initial erosion would then accelerate the wall degradation by carving deep grooves into the thermal liner through direct particle impingement and spallation (Fig. 40).

7. Conclusion

The injector design methodology developed in this work was applied to produce two injector variants for the ABLE technology demonstrator engine, with an ideal thrust of 18 kN at a chamber pressure of 12 bar. Cold-flow testing validated the method's capacity to achieve target flow uniformity: Injector A met the design specification of $\pm 5\%$ flow distribution, confirming that the methodology can produce a balanced injector layout when manufacturing quality is maintained. In contrast, Injector B exhibited asymmetric flow profiles due to geometric imperfections, specifically burrs at the orifice inlets and poor orifice tolerances, highlighting the sensitivity of injector performance to fabrication fidelity, even when guided by a robust design process.

Despite these imperfections, Injector B underwent successful hot-fire tests, demonstrating stable combustion and achieving nominal efficiency. However, a long-duration test on a compromised ablative liner revealed excessive erosion. The high momentum ratios between jet pairs is theorized to have created misalignment between adjacent spray fans, resulting in localized oxidizer-rich zones. While this identifies a limitation in the current design's sensitivity to manufacturing tolerances, the engine's overall stability validates the methodology's emphasis on conservative, high-stability configurations.

Overall, the methodology provided a sound injector architecture, ensuring safe and effective engine operation across all tests. The ABLE engine's successful demonstration validates the design approach's effectiveness in producing reliable, first-iteration liquid rocket engine injectors, even within the manufacturing constraints of a developing country. This structured methodology offers a valuable blueprint for emerging space programs seeking indigenous rocket engine development capabilities.

Funding

This work was sponsored by the Department of Science, Technology and Innovation and the South African National Space Agency.

CRediT authorship contribution statement

Duran Martin: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Michael Brooks:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization; **Jean Pitot:** Supervision, Project administration, Funding acquisition, Conceptualization; **Ryan Cooper:** Methodology, Investigation, Conceptualization; **Nino Wunderlin:** Investigation, Conceptualization.

Data availability

The authors do not have permission to share data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was sponsored by the Department of Science, Technology and Innovation and the South African National Space Agency. The authors would like to express their gratitude to the DSTI Space and Technology Directorate for their support and to CSIR's Centre for High Performance Computing for the computational resources.

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